



Basic Principles of Medicine 1

Module: Musculoskeletal System | Lecture 04

Clinical Anatomy of the Axilla & Brachial Plexus

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Recommended Reading

- Drake, Vogl, Mitchell. Grays Anatomy for Students. 4th ed. Elsevier Churchill Livingstone.
- Printed Textbook: Chapter 7
- Online Textbook: Chapter 7: [click here](#)
 - Conceptual Overview
 - Regional Anatomy:
 - Posterior scapular region
 - Axilla
 - Brachial plexus
- Pay special attention to the **green (printed text)** / **blue (ebook)** “*In the Clinic*” boxes for the clinical correlations for this topic

Objectives

SOM.MK.II.BPM1.1.MSK.1.ANAT.1122

Describe the location of the axilla and the structures that form its medial, lateral, anterior and posterior boundaries (walls), and its apex.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1123

Describe the cervicoaxillary sheath and its clinical importance.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1124

Describe the clinical significance of the axillary sheath in local anesthesia.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1125

Describe the subclavian artery and its branches, based on what they supply and where they are located.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1126

List the general attachments, innervations and actions of the muscles that contribute to the walls of the axilla.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1127

Describe the functional loss of the muscles of the axillary walls resulting from lesions to the nerves innervating these muscles.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1128

Describe the formation of the axillary vein and the major tributaries that join it.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1129

Describe the division of the axillary artery into its three parts and the structures related to each part.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1130

List the arteries involved in anastomosis around the scapula, give the origin artery for each and discuss the clinical importance of this anastomoses.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1131

Describe the lymphatic drainage of the walls of the axilla use this information to interpret clinical signs and symptoms

SOM.MK.II.BPM1.1.MSK.1.ANAT.1132

Describe the organization of the brachial plexus from its roots to its terminal branches.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1133

Define the terms pre-fixed and post-fixed brachial plexus and discuss its clinical importance.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1134

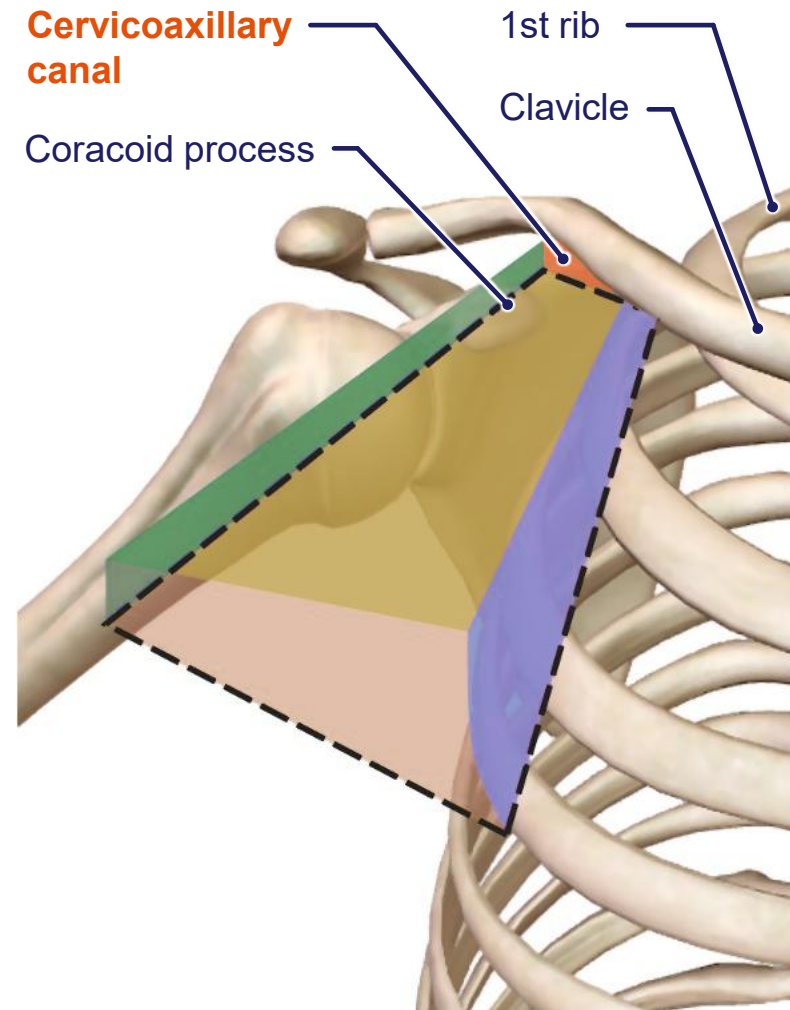
List the fiber types of the nerves that are present in the branches of the brachial plexus.

SOM.MK.II.BPM1.1.MSK.1.ANAT.1135

Describe the anatomical relationships that are important for successful insertion of a catheter for central venous access (infraclavicular subclavian venipuncture) (**Grey's Anatomy "In the Clinic" box**).

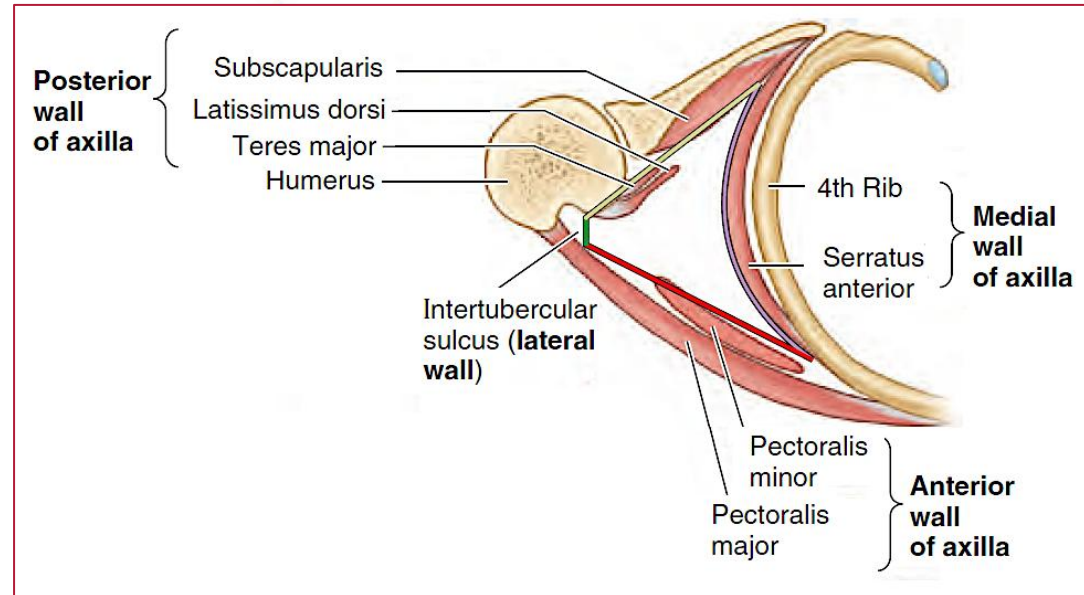
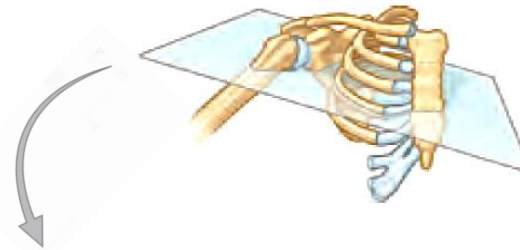
Axilla

- Pyramidal-shaped space between upper arm and thorax
- Superior portion (**apex**) forms a passageway to the neck region
 - also known as axillary inlet or cervicoaxillary canal
 - found between:
 - clavicle
 - coracoid process and superior margin of scapula
 - 1st rib
- Axilla bounded by floor and 4 walls:
 - Anterior
 - **Posterior**
 - **Medial**
 - **Lateral**



Boundaries

- **Anterior (anterior fold):**
 - Pectoralis major and minor
- **Posterior (posterior fold):**
 - Teres major, latissimus dorsi, subscapularis
- **Medial:**
 - Serratus anterior, ribs
- **Lateral:**
 - Intertubercular (bicipital) groove of humerus
- **Floor:**
 - Axillary fascia



Muscles of the Anterior Wall

Pectoralis major

- Flexion, **adduction**, and medial rotation of the arm
- **Medial and lateral pectoral nerves**

Pectoralis minor

- Pulls tip of shoulder down
- **Medial pectoral nerve**

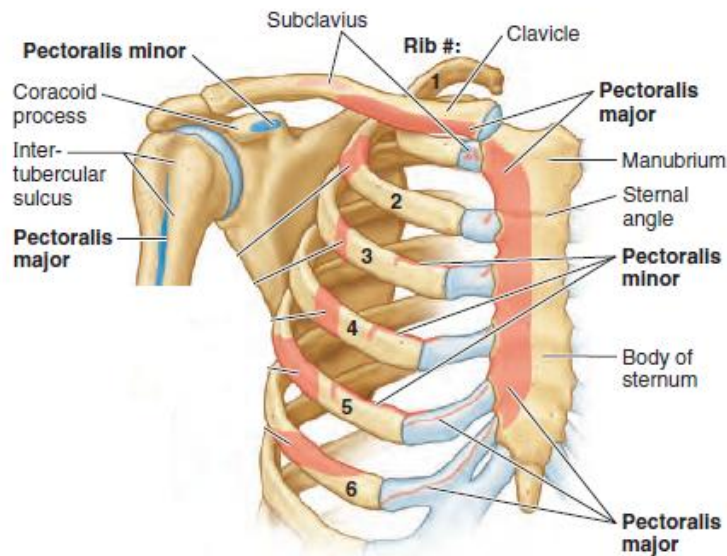
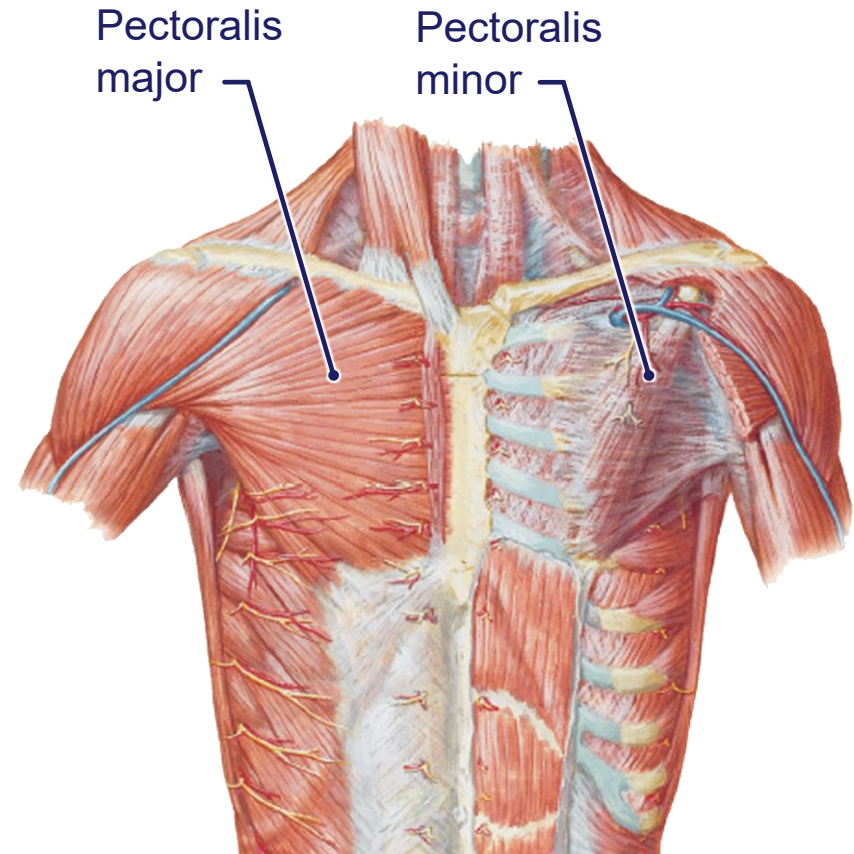


FIGURE 6.21



Muscles of the Posterior Wall

Subscapularis

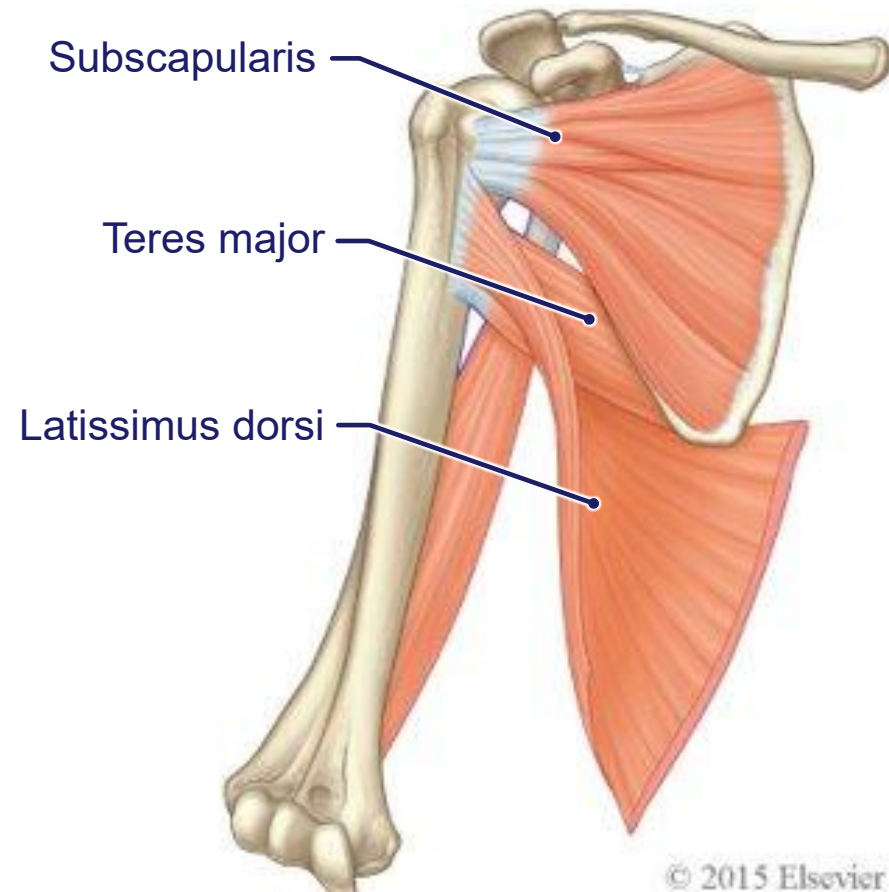
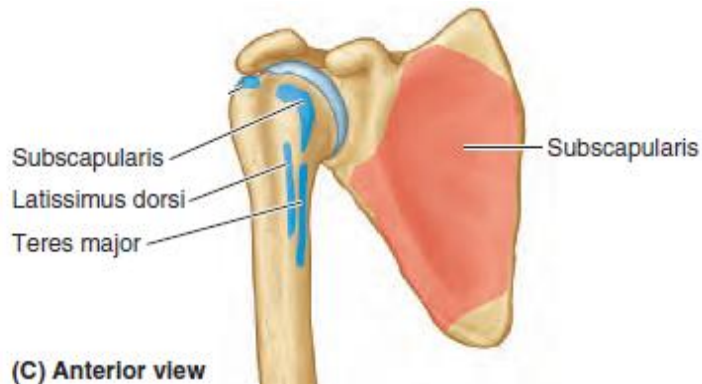
- Medial rotation of the arm
- Upper and lower subscapular nerves

Teres major

- Medial rotation, **adduction** and extension of the arm
- Lower subscapular nerve

Latissimus dorsi

- Medial rotation, **adduction** and extension of the arm
- Thoracodorsal nerve



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Muscles of the Posterior Wall

Subscapularis

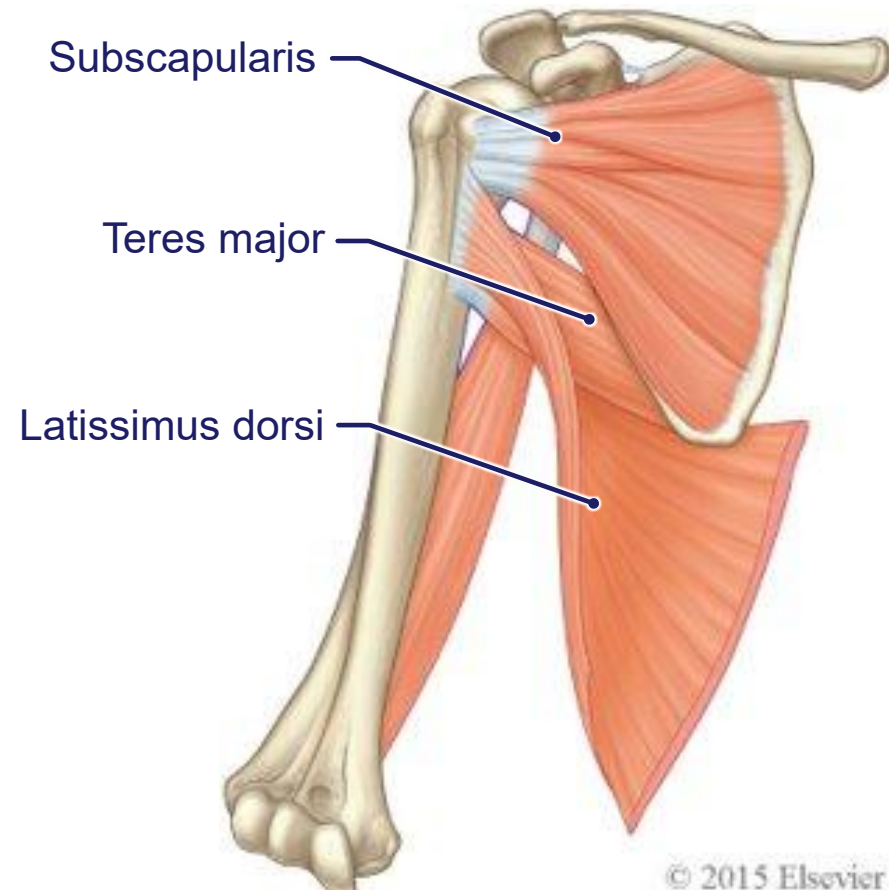
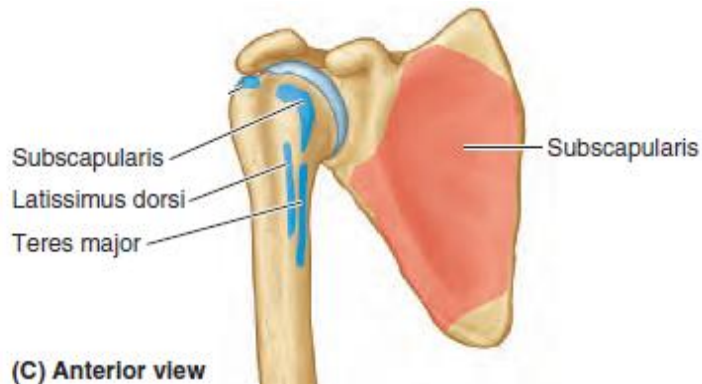
- Medial rotation of the arm
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Teres major

- Medial rotation, **adduction** and extension of the arm
- Lower subscapular nerve

Latissimus dorsi

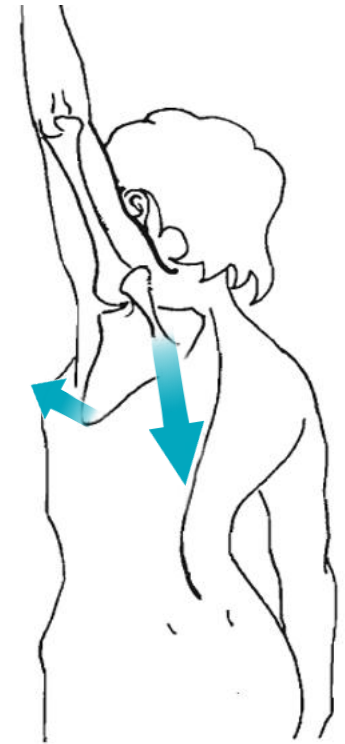
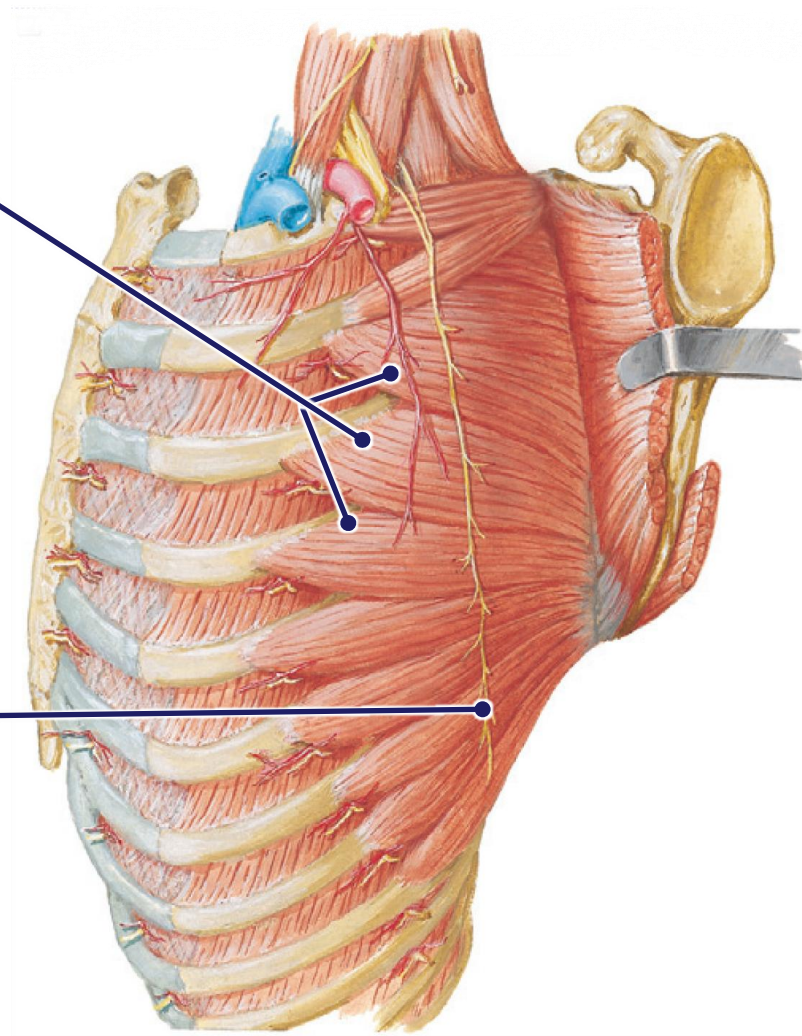
- Medial rotation, **adduction** and extension of the arm
- Thoracodorsal nerve



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Muscles of the Medial Wall

- Serratus anterior**
 - Rotates scapula upward to allow lifting arm above 90°
 - Protracts the scapula to keep the medial border against the thoracic wall
- Long thoracic nerve (C5,6,7)**



Clinical Correlate: Winged Scapula

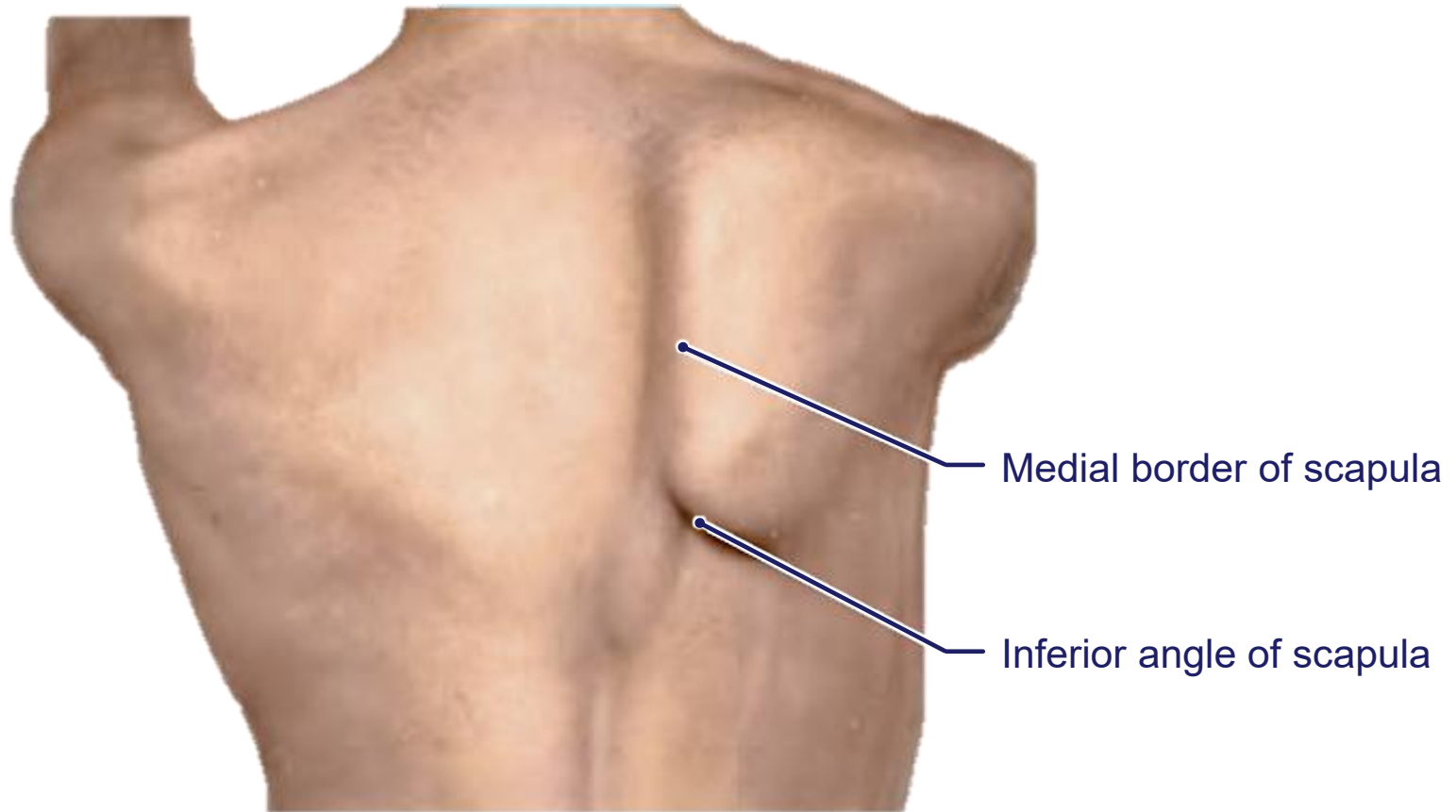
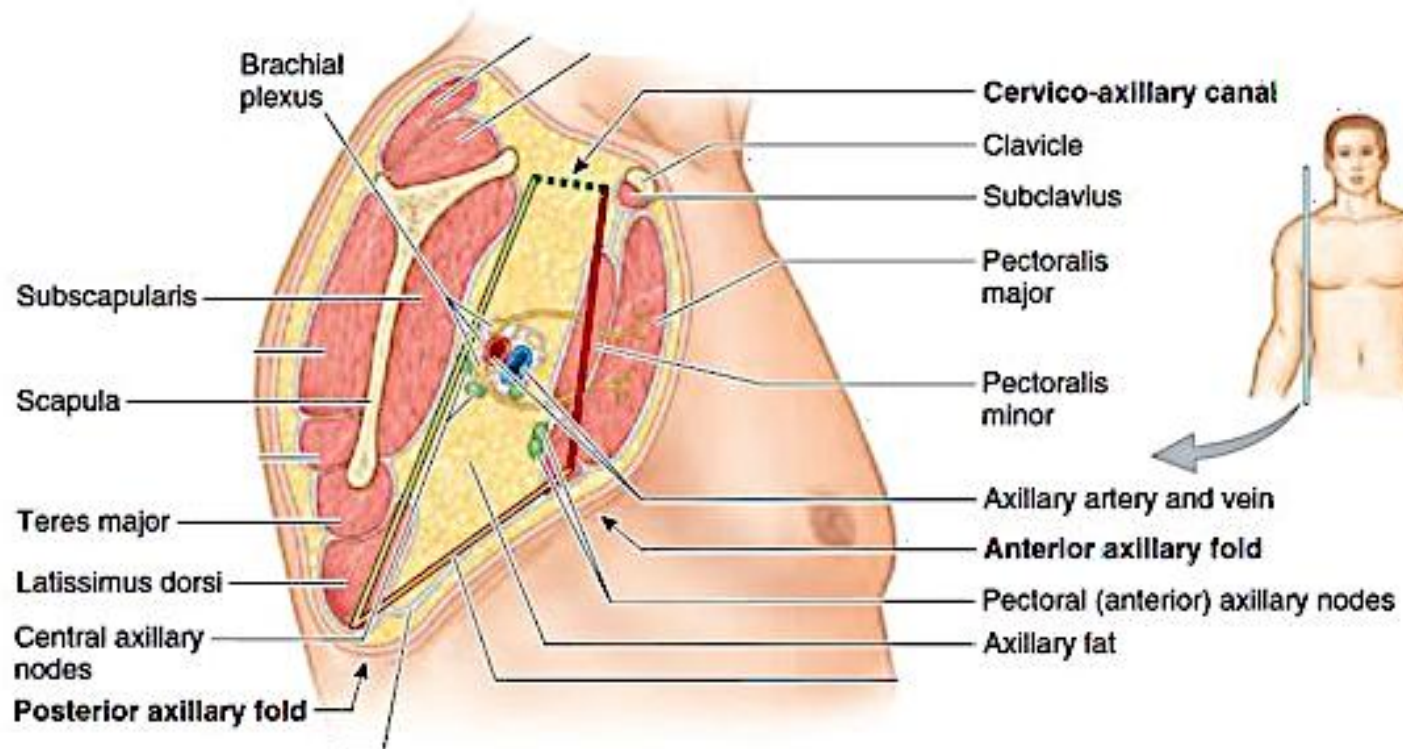


FIGURE B6.5. Right winged scapula

Contents of Axilla

- **Vessels**
 - Axillary artery and vein
- **Nerves**
 - Cords of brachial plexus: lateral, posterior, medial
- **Lymph nodes and fat**

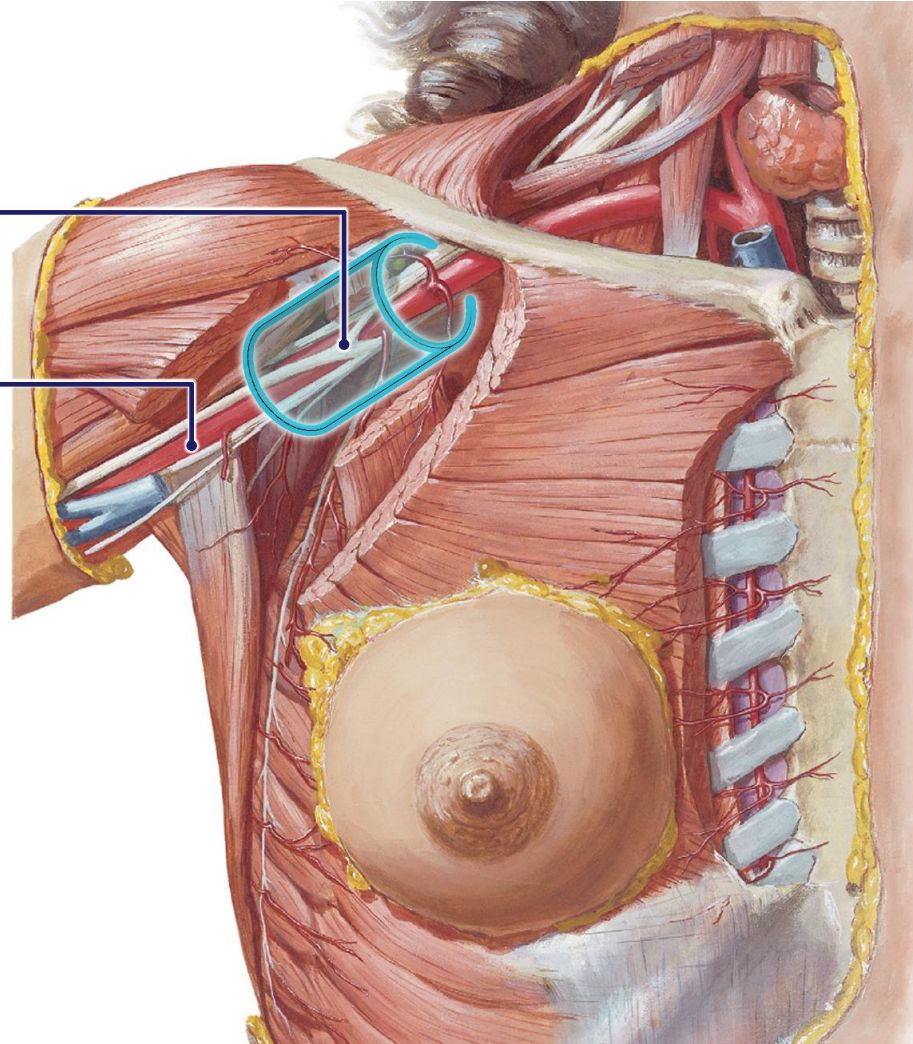


Cervicoaxillary Canal

- Passageway for the neurovascular bundle from the neck to the upper limb and vice versa

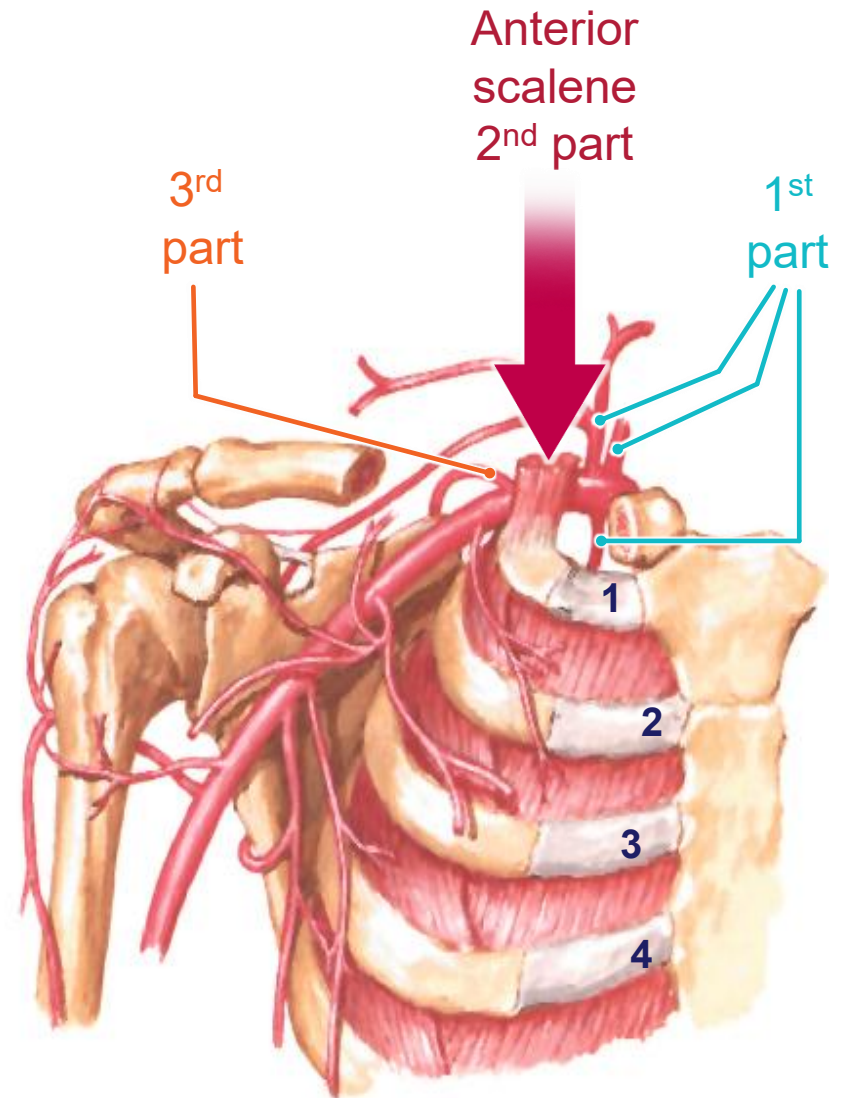
Brachial plexus

Axillary artery



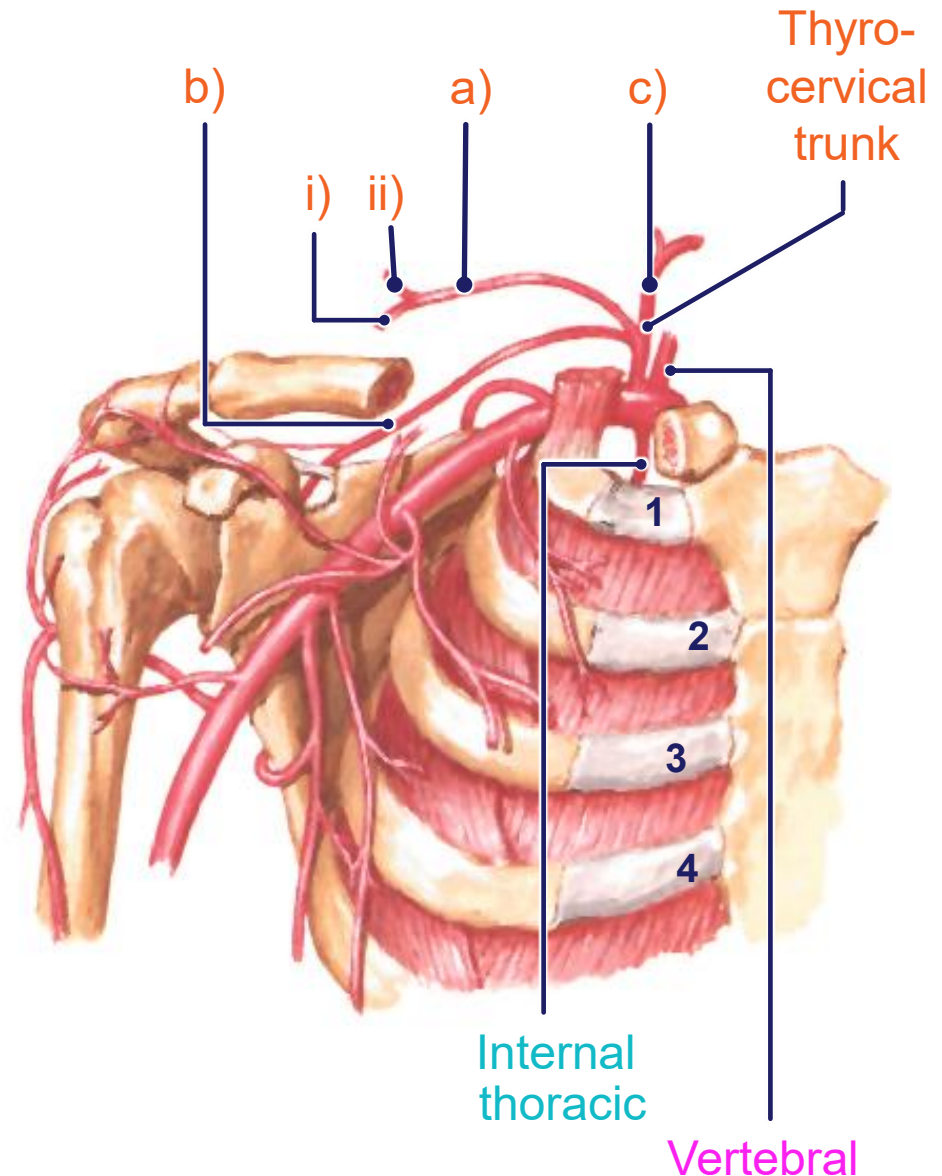
Subclavian artery

- Divided into 3 parts by the anterior scalene muscle
- 1st part
 - medial to anterior scalene
- 2nd part
 - Posterior to anterior scalene
- 3rd part
 - Lateral to anterior scalene



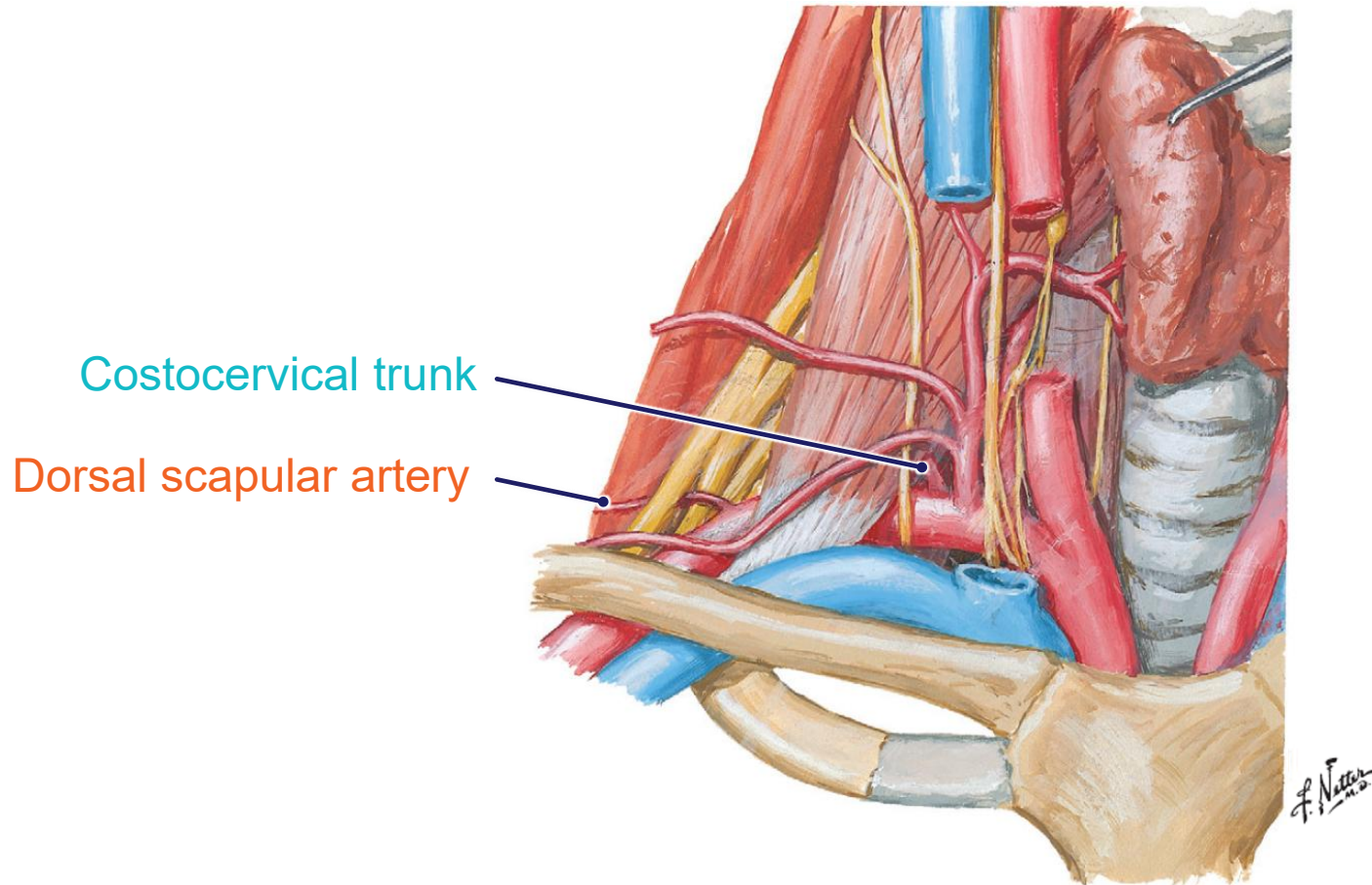
Subclavian artery – 1st part

- Most branches from the first division
- 1st part (medial to scalene)
 - Vertebral
 - Internal thoracic
 - Thyrocervical trunk
 - a) Transverse cervical
 - i) Deep
 - ii) Superficial
 - b) Suprascapular
 - c) Inferior thyroid



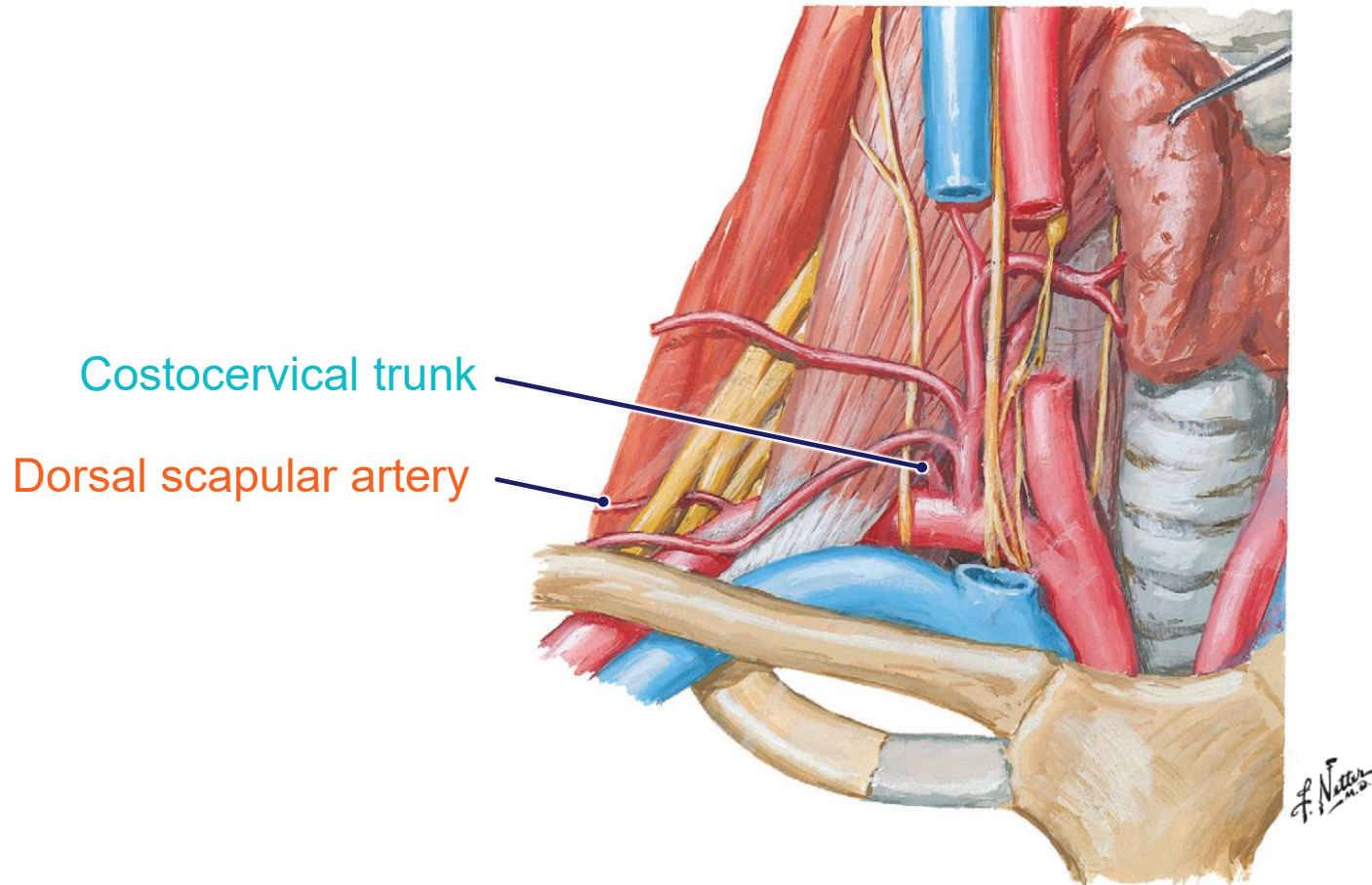
Subclavian artery – 2nd & 3rd parts

- 2nd part (posterior to scalene): **Costocervical**
- 3rd part (lateral to scalene): **Dorsal scapular (if present)**



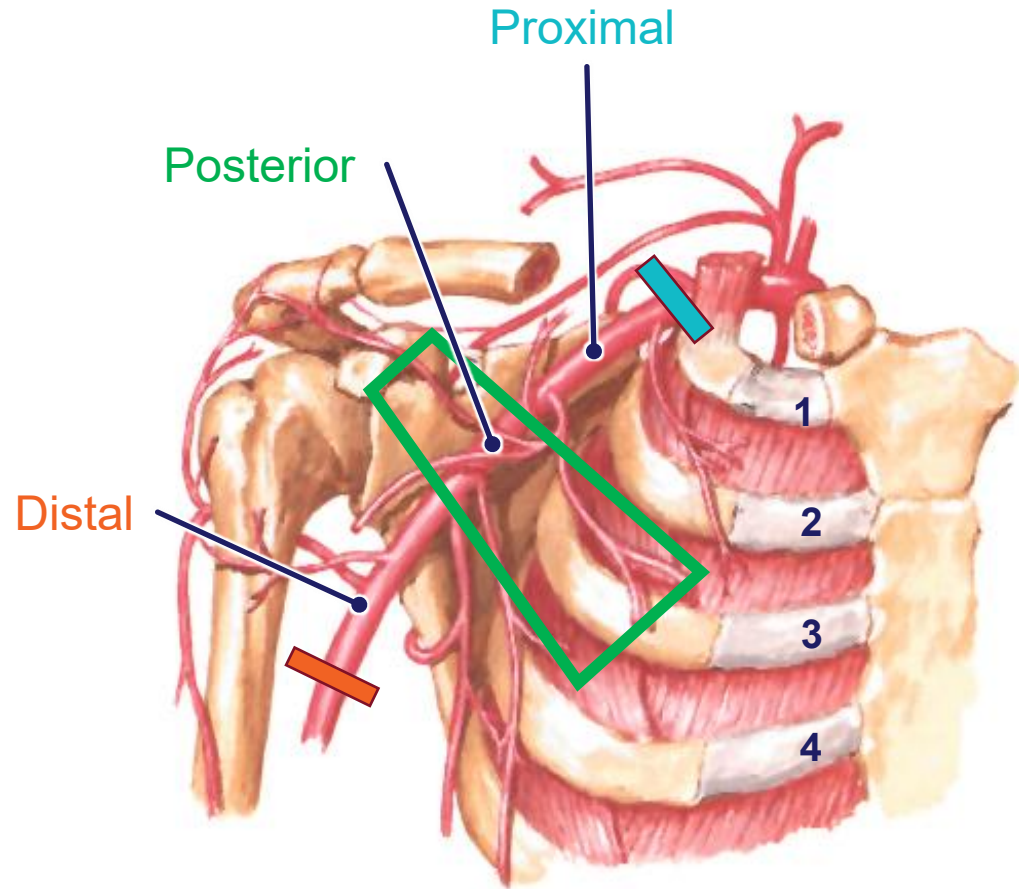
Subclavian artery – 2nd & 3rd parts

- 2nd part (posterior to scalene): **Costocervical**
- 3rd part (lateral to scalene): **Dorsal scapular (if present)**



Axillary artery

- Continuation of subclavian artery
- Begins at 1st rib lateral border (proximal)
- Ends at inferior margin of teres major (distal)
- Divided into 3 parts by pectoralis minor
 - Proximal
 - Posterior
 - Distal

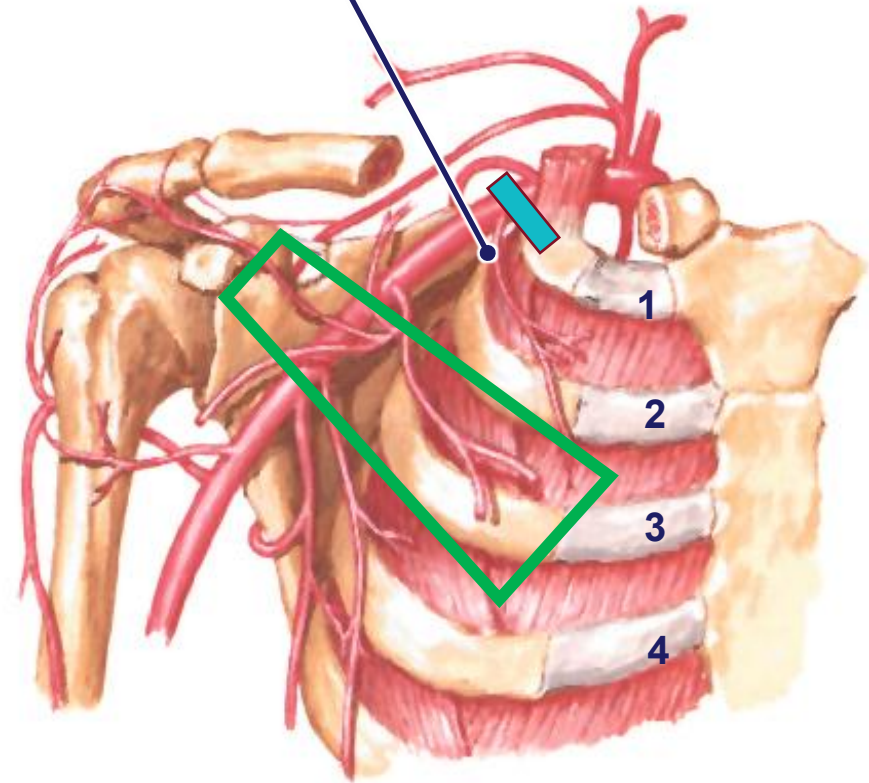


Axillary artery – proximal part

1st Part: Proximal

- Superior thoracic

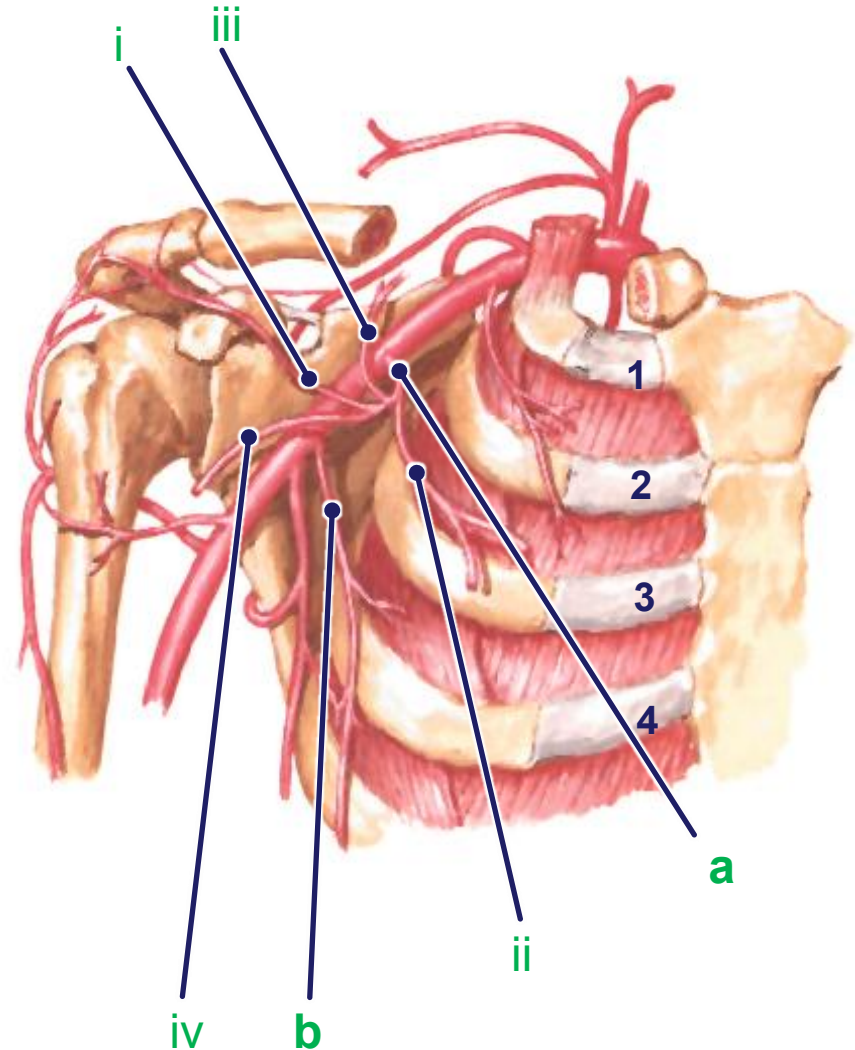
Superior thoracic



Axillary artery – posterior part

2nd Part: Posterior

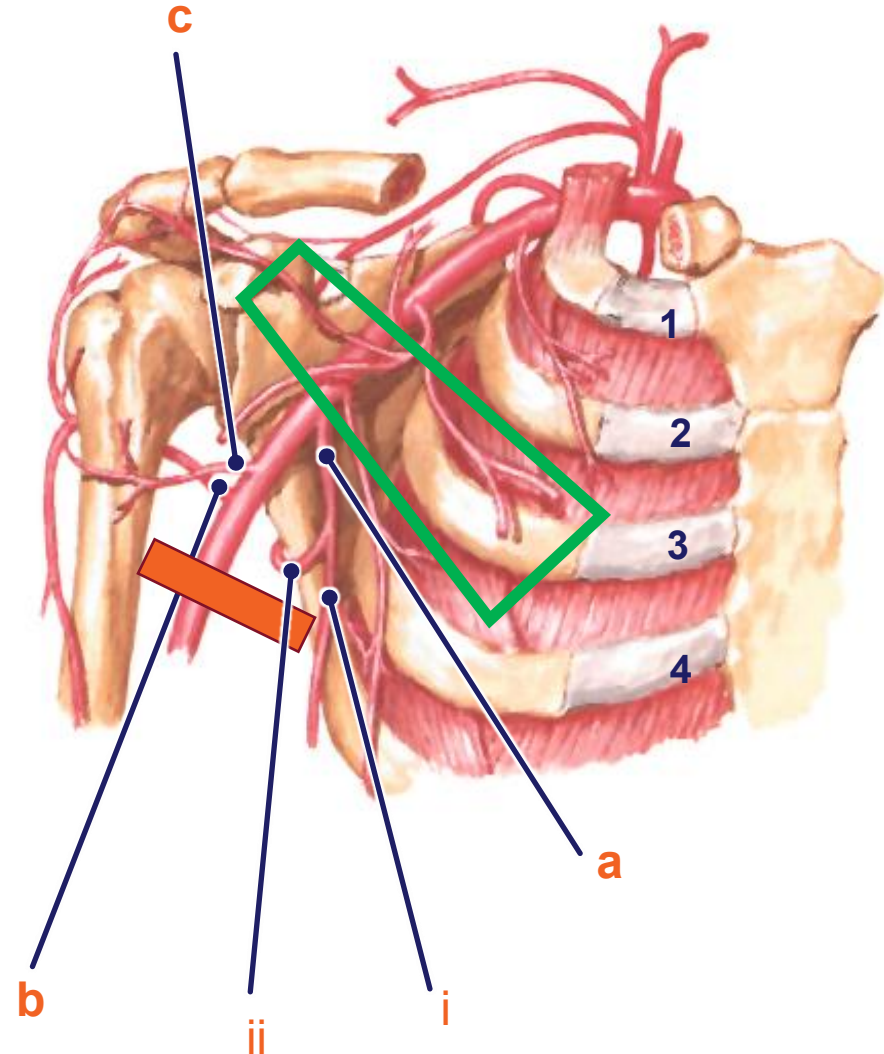
- **a) Thoracoacromial trunk**
 - i) acromial
 - ii) pectoral
 - iii) clavicular
 - iv) deltoid
- **b) Lateral thoracic**



Axillary artery – distal part

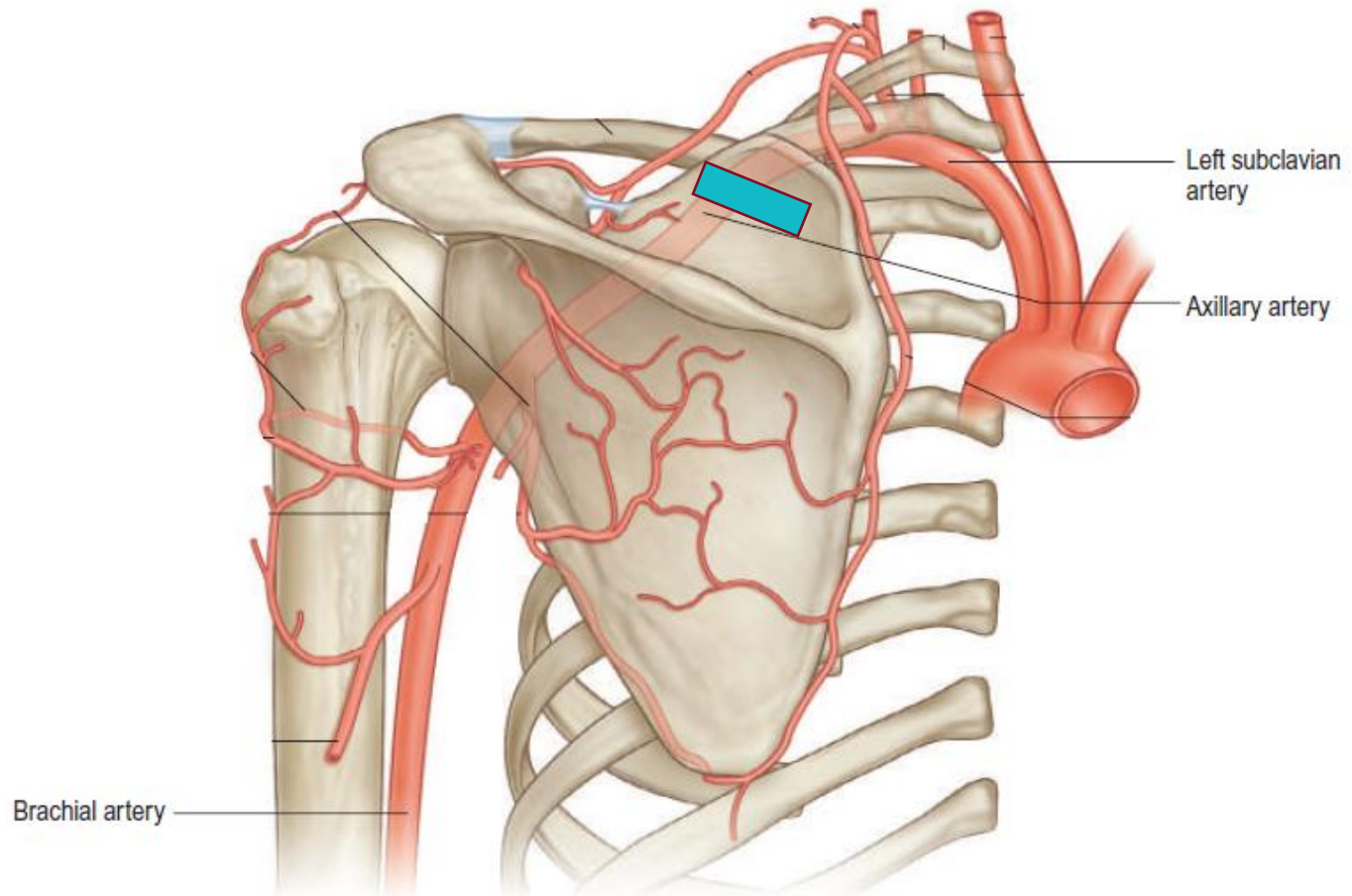
3rd Part: Distal

- **a)** Subscapular
 - i) thoracodorsal
 - ii) circumflex scapular
- **b)** Posterior circumflex humeral
- **c)** Anterior circumflex humeral



Blood flow to the arm and forearm

What happens if there is ligation (blockage) of the axillary artery?

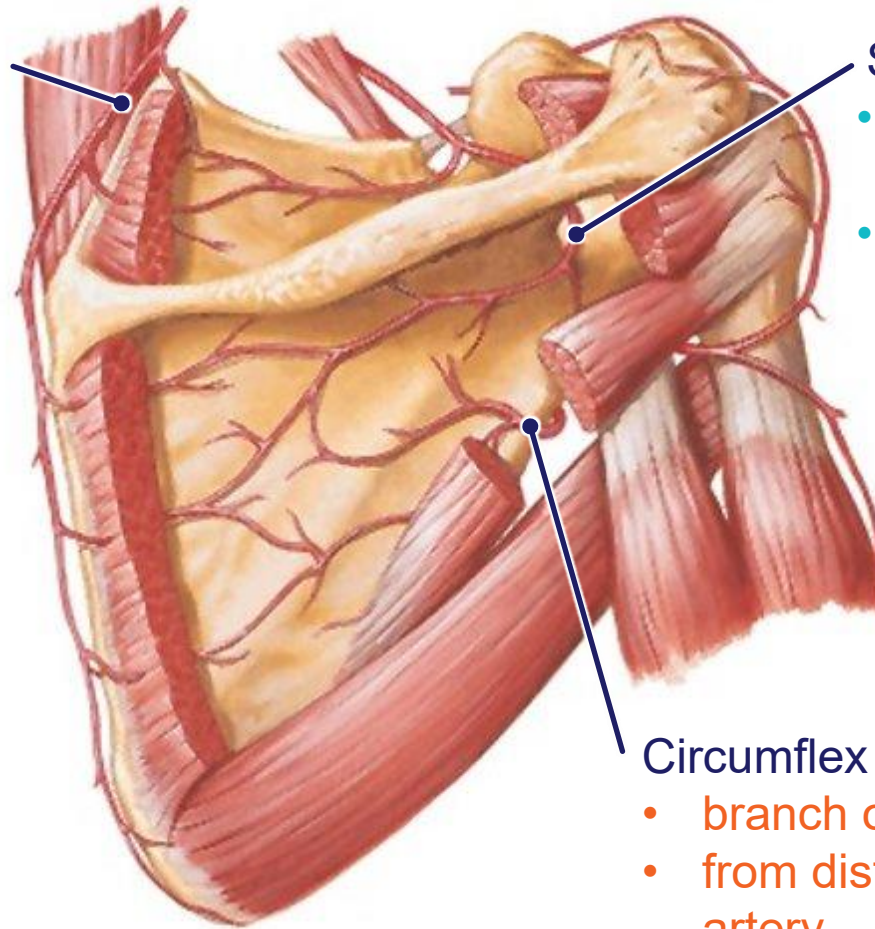


Standring et al., *Gray's Anatomy*, 41st ed, 2016

Scapular anastomoses

Deep branch of
transverse cervical
or
Dorsal scapular

- branch of thyrocervical trunk
- from subclavian artery



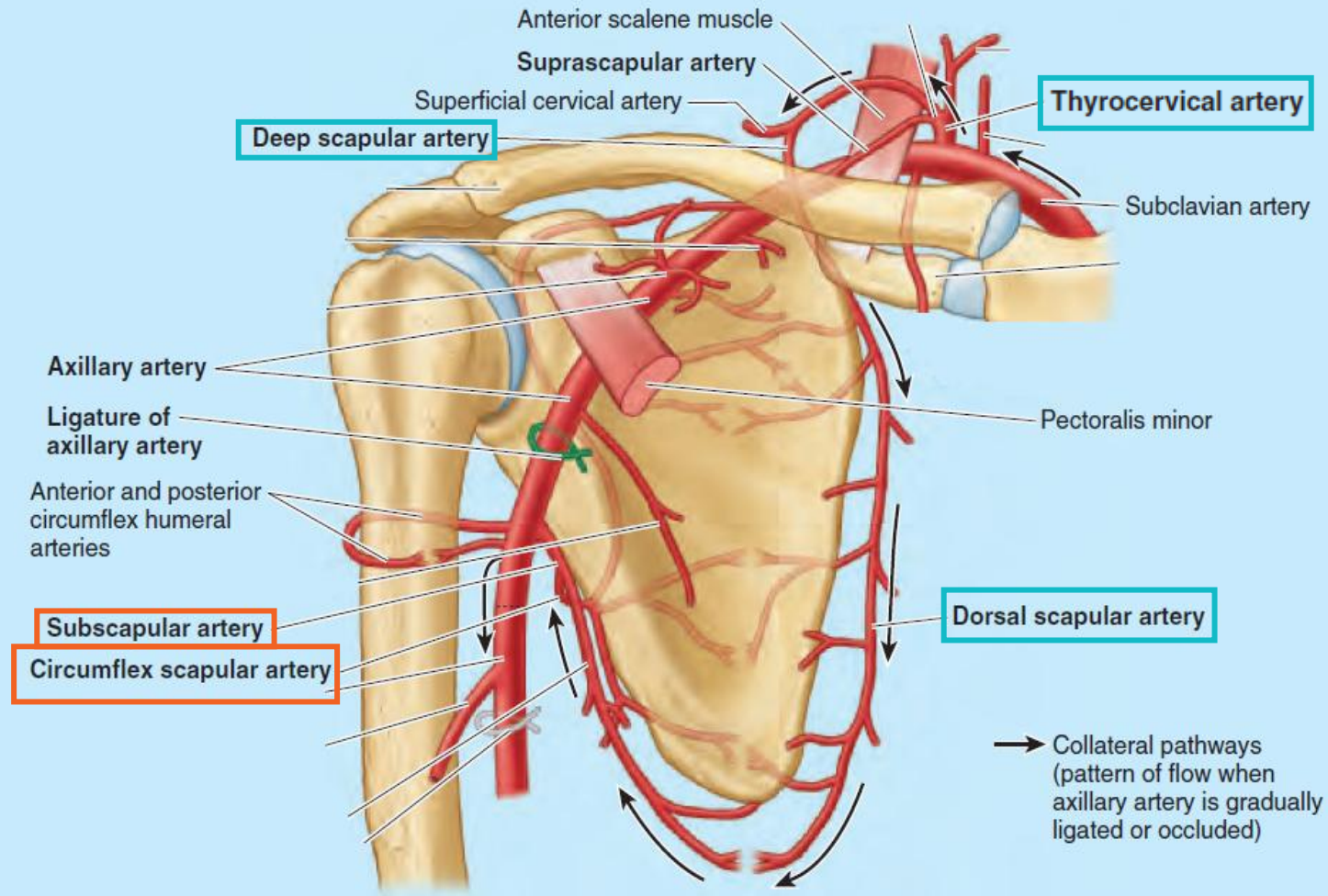
Suprascapular artery

- branch of thyrocervical trunk
- from subclavian artery

Circumflex scapular artery

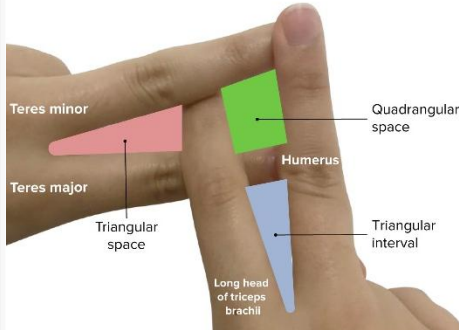
- branch of subscapular artery
- from distal part of axillary artery

Scapular collateral pathways



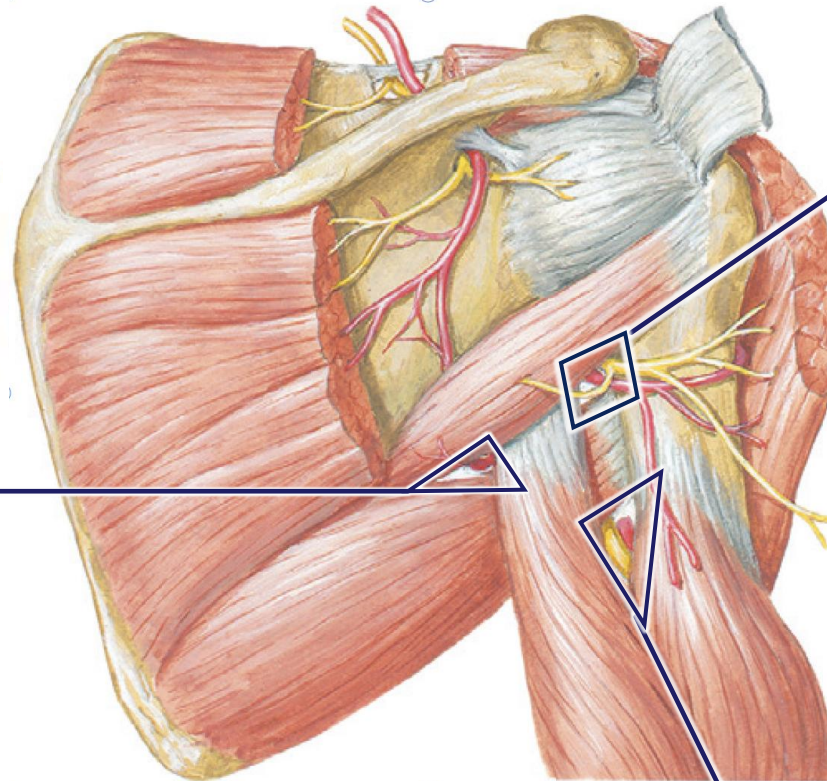
Posterior Scapular Spaces

Axillary spaces using finger mnemonic



Triangular space

- circumflex scapular artery



Quadrangular space

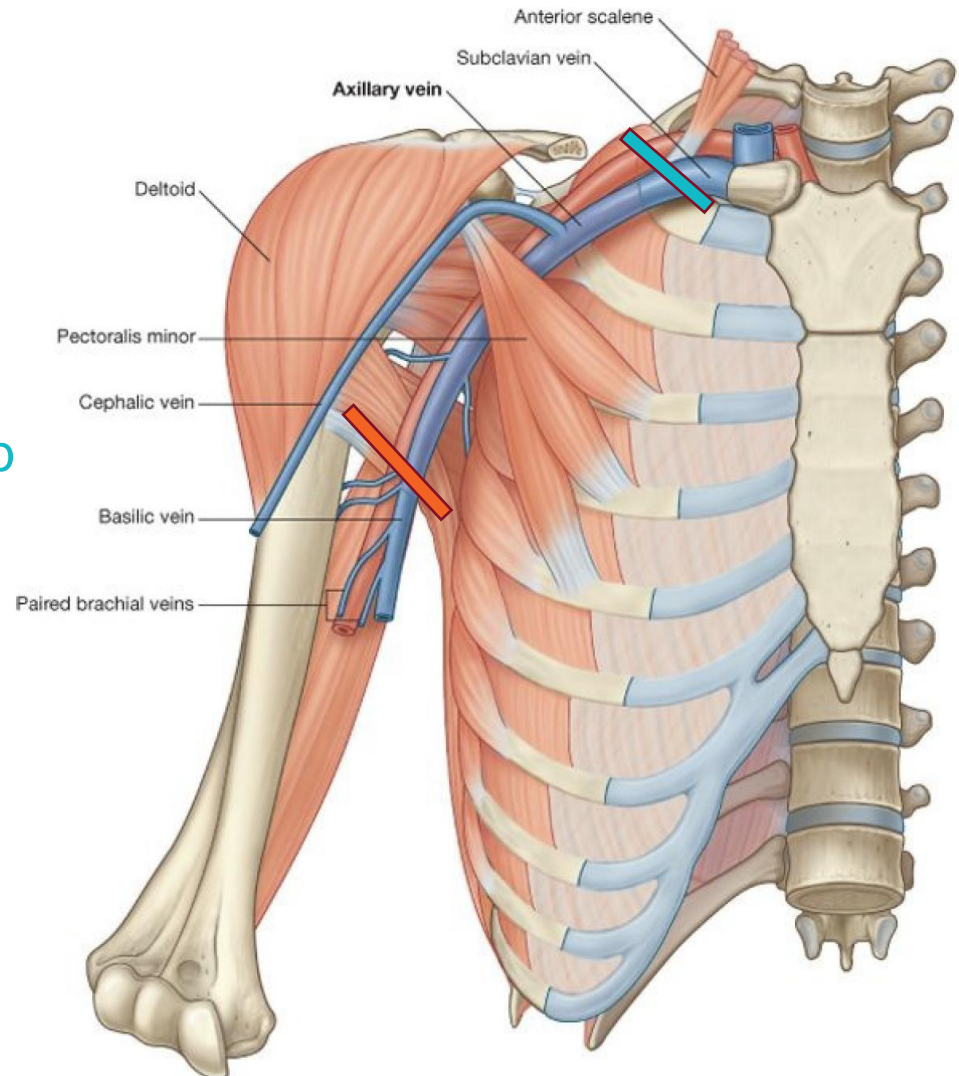
- axillary nerve
- posterior humeral circumflex artery

Triangular interval

- radial nerve
- deep brachial artery

Axillary vein

- Formed by basilic vein and paired brachial veins
- Begins at inferior margin of teres major
- Ends at lateral border of 1st rib
- Receives cephalic vein

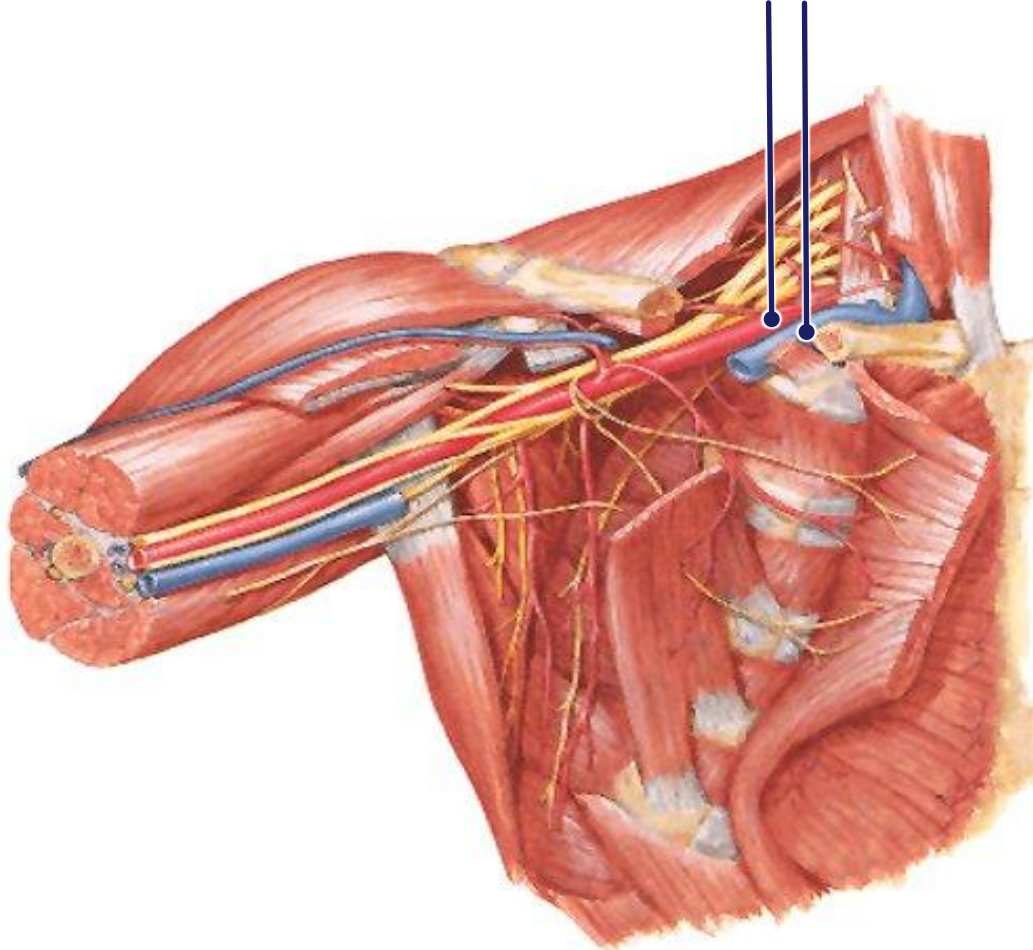


Drake: Gray's Anatomy for Students, 2nd Edition.

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Clinical Correlate: Axillary artery and vein

Axillary artery and vein



- Axillary vein may be used for entry into the subclavian vein
- Used during insertion of a ***subclavian venous catheter***

Lymphatic Drainage of the Axilla

Pectoral (anterior) nodes

- Receive lymph from antero-lateral thoracic wall and breast

Subscapular (posterior) nodes

- Receive from posterior thoracic wall, posterior shoulder, and back

Humeral (lateral) nodes

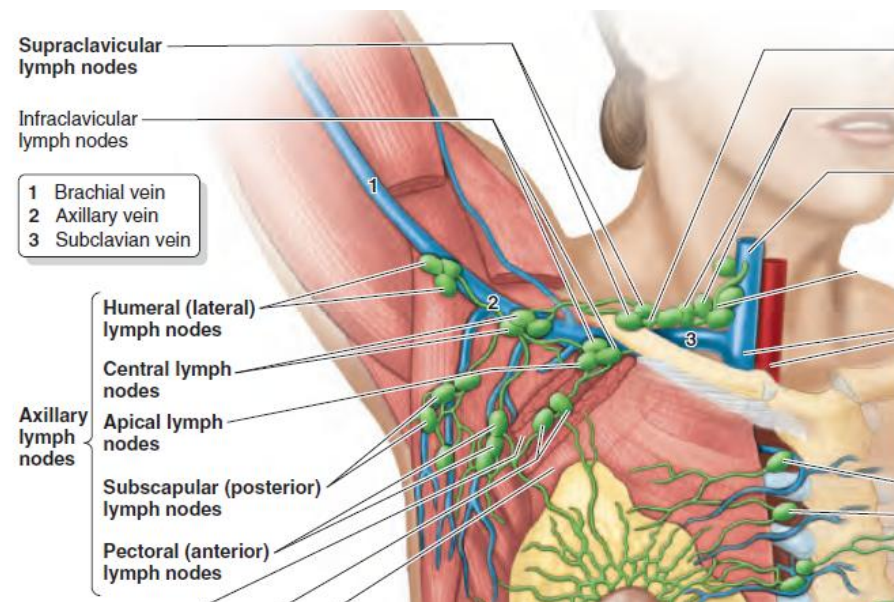
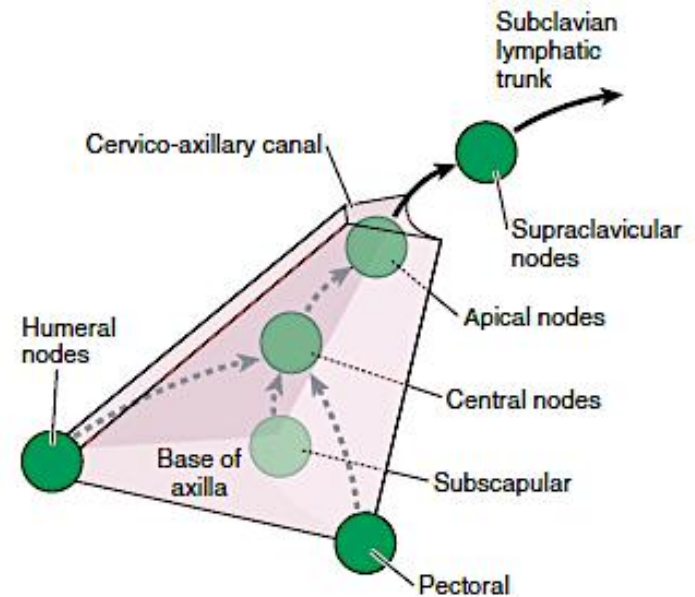
- Receive from **medial aspect of upper limb** accompanying the basilic vein

Central nodes

- Receive from anterior, lateral, and posterior groups

Apical nodes

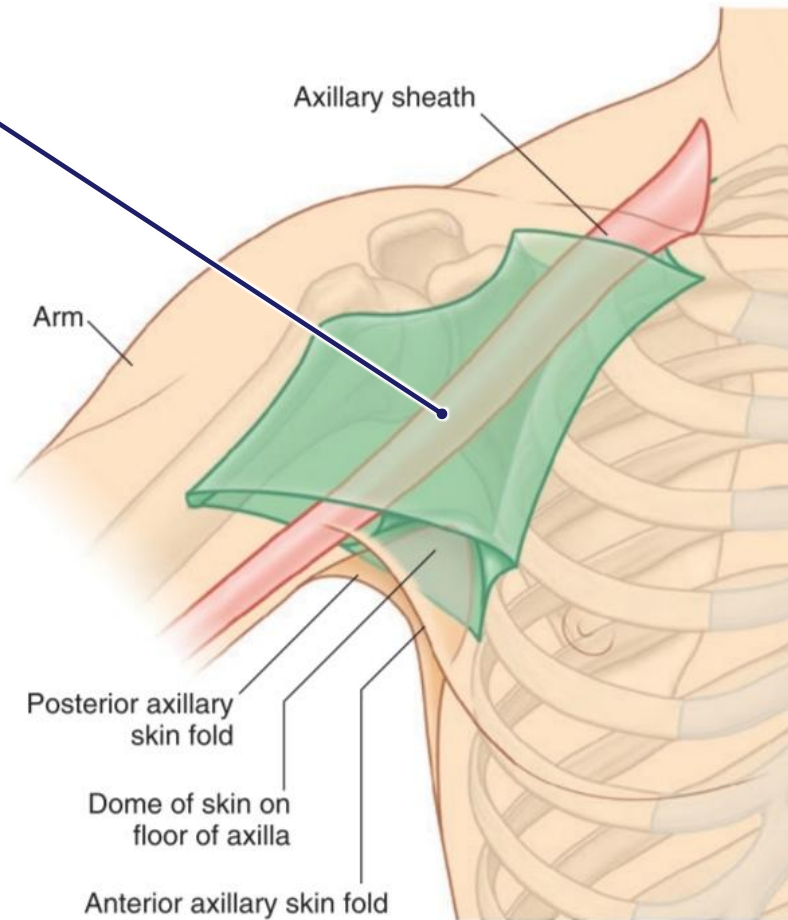
- Receive from all other axillary nodes and superficial lymphatics from **lateral aspect of upper limb** accompanying cephalic vein



Clinical Correlate: Axillary sheath

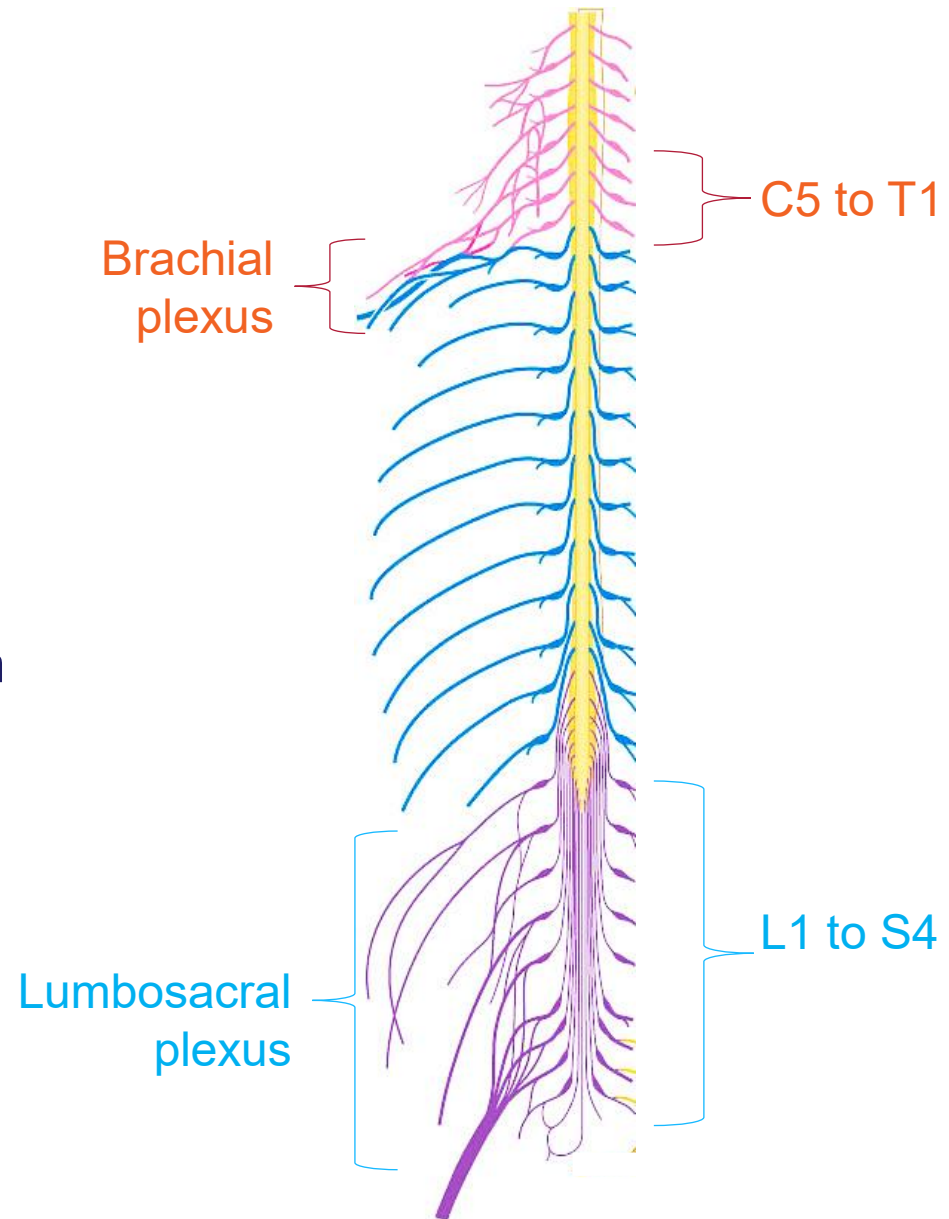
Axillary sheath

- Surrounds arteries, veins, nerves and lymphatics
- Can be used for ***brachial plexus blocks***

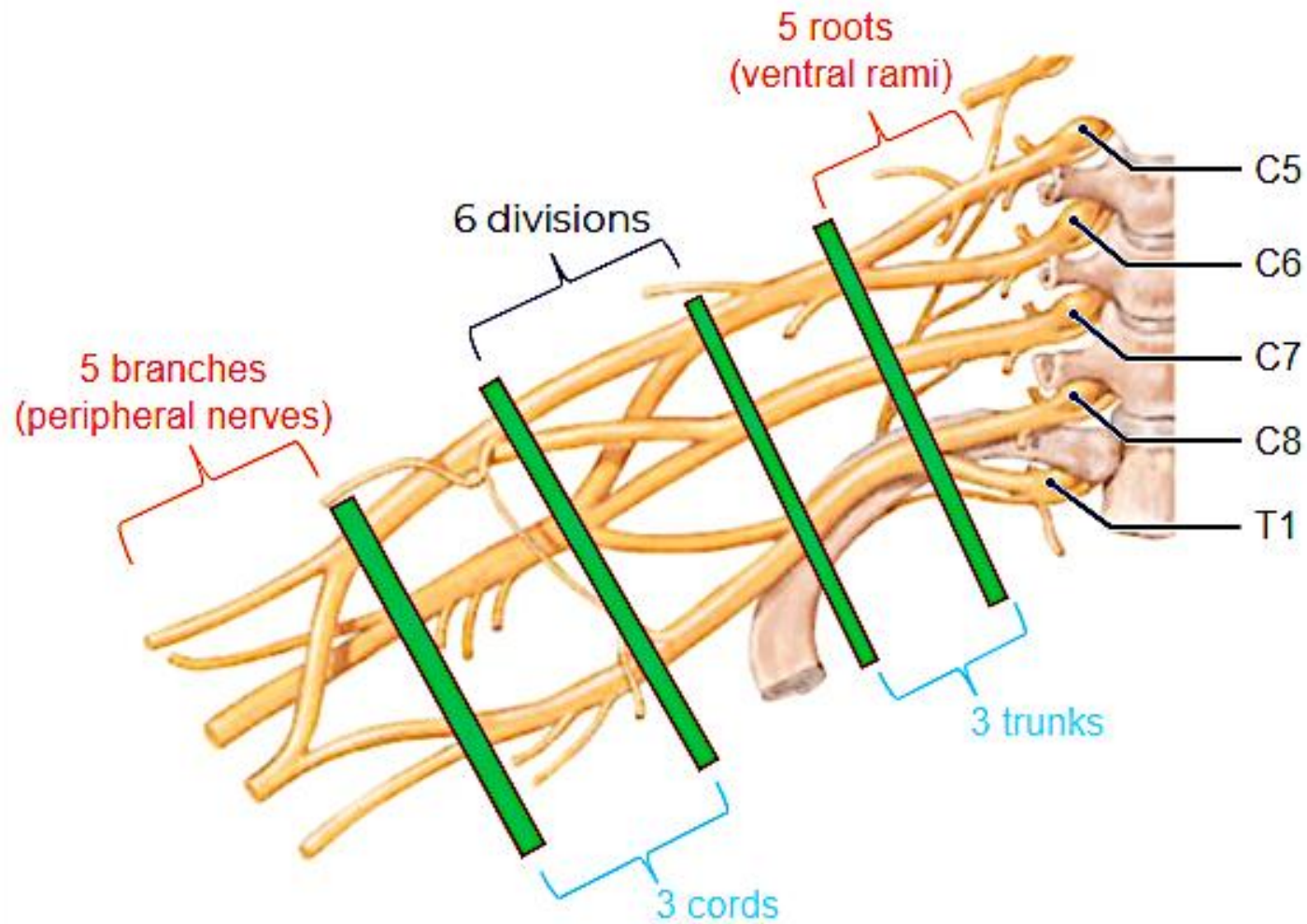


Somatic Plexus

- Thoracic ventral rami innervating the trunk remain separate from each other
 - Ventral rami innervating the limbs merge with adjacent rami to form somatic plexuses
 - Nerve fibers from adjacent rami intermingle
- Named nerves arise from plexus and contain **fibers from two or more rami**

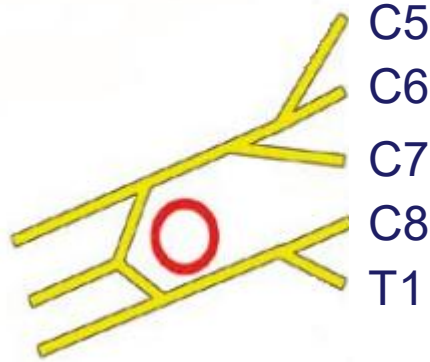


Brachial Plexus Overview

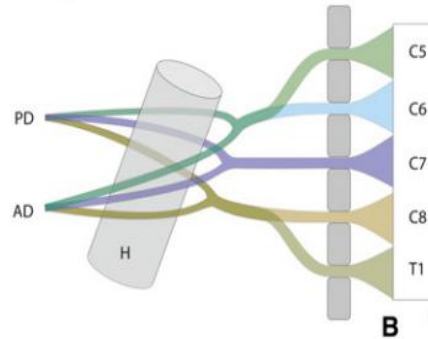


Brachial Plexus Organization

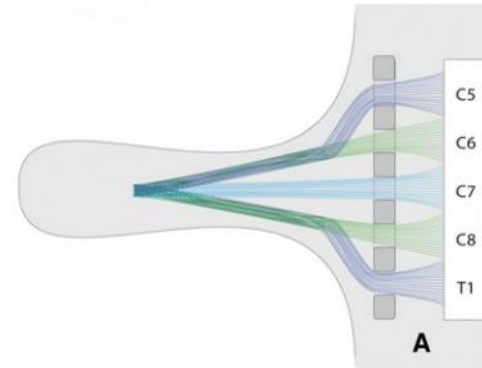
Cords encircle the axillary artery before ending in branches



Anterior and Posterior division form around skeleton precursors

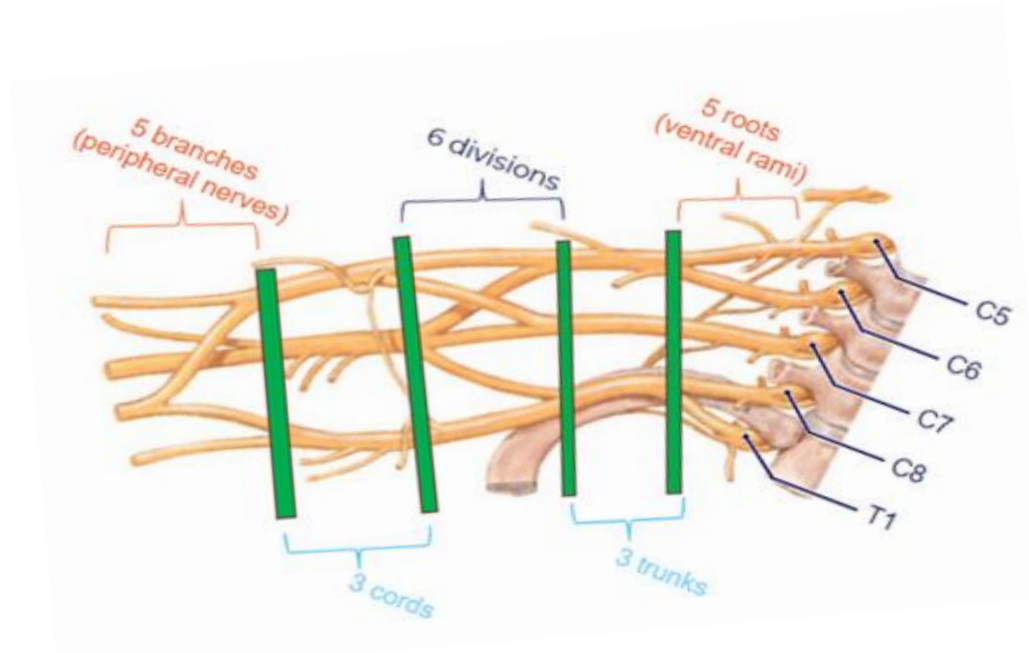


Roots converge into trunks in the developing limb bud



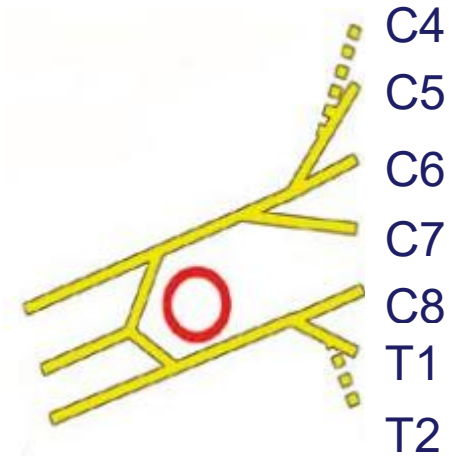
Kawashima et al., Folia Morphol 81(4):884-899, 2022

Leijnse et al., J Anat 236:862-882, 2020

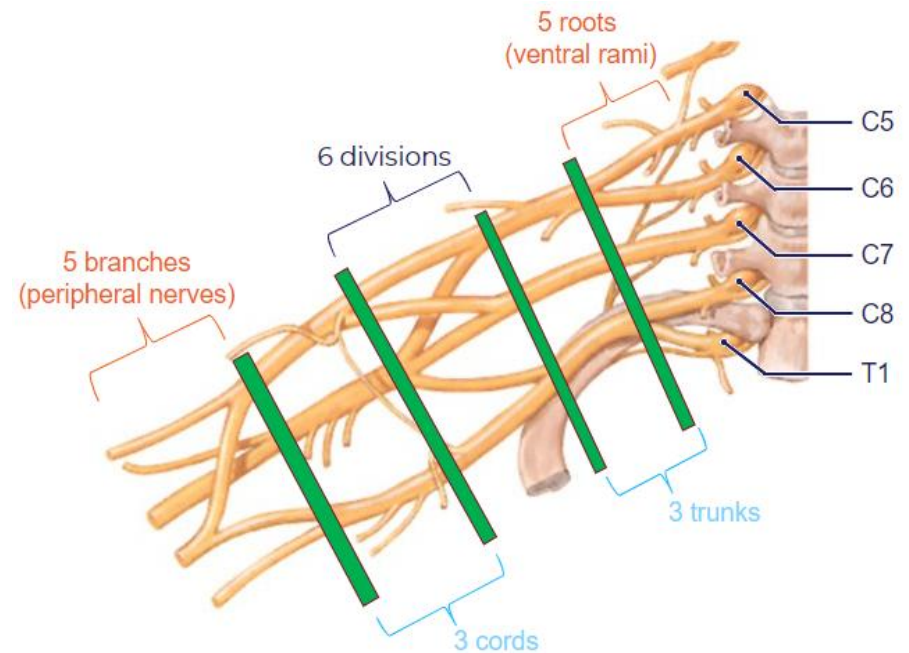


Brachial Plexus Contributions

- Formation of the brachial plexus can lead to variations
- Small contributions may be made in addition to the typical ventral rami:
- May be **pre-fixed** (C4-C8)
 - Superiormost root is C4
 - Inferiormost root is C8
- May be **post-fixed** (C6-T2)
 - Superiormost root is C6
 - Inferiormost root is T2



Kawashima et al., Folia Morphol 81(4):884-899, 2022



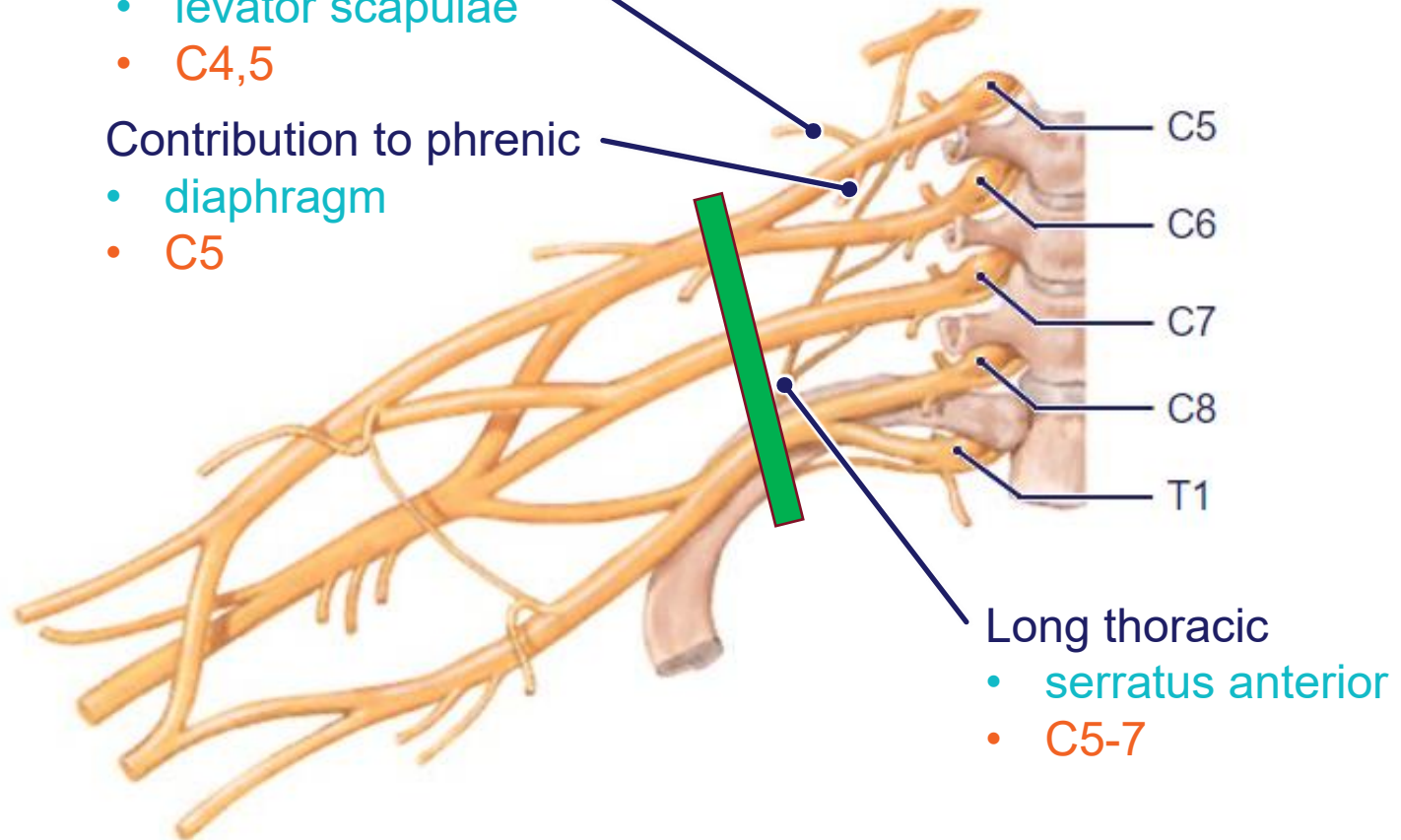
Brachial Plexus: 5 Roots

Dorsal scapular

- rhomboids
- levator scapulae
- C4,5

Contribution to phrenic

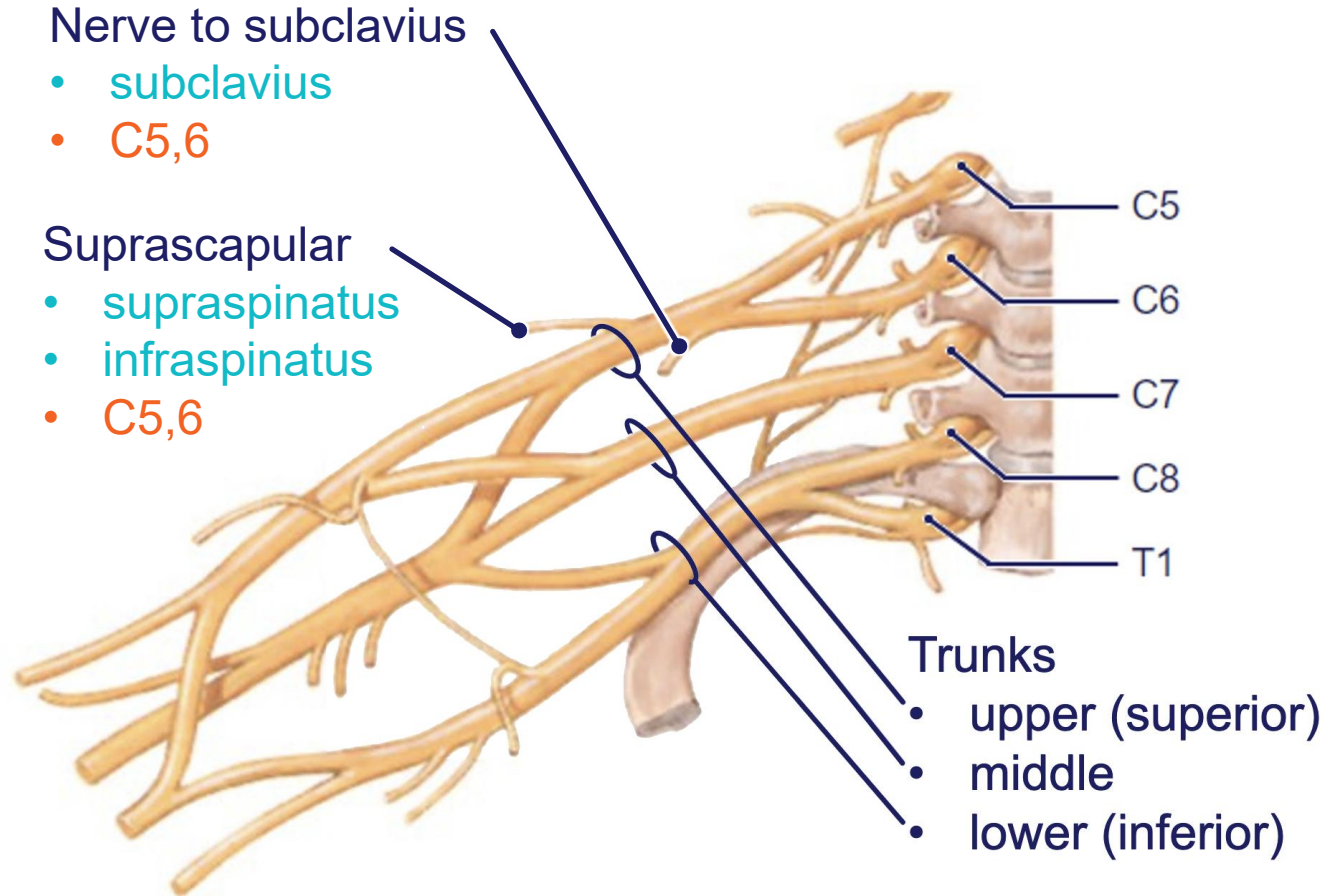
- diaphragm
- C5



Long thoracic

- serratus anterior
- C5-7

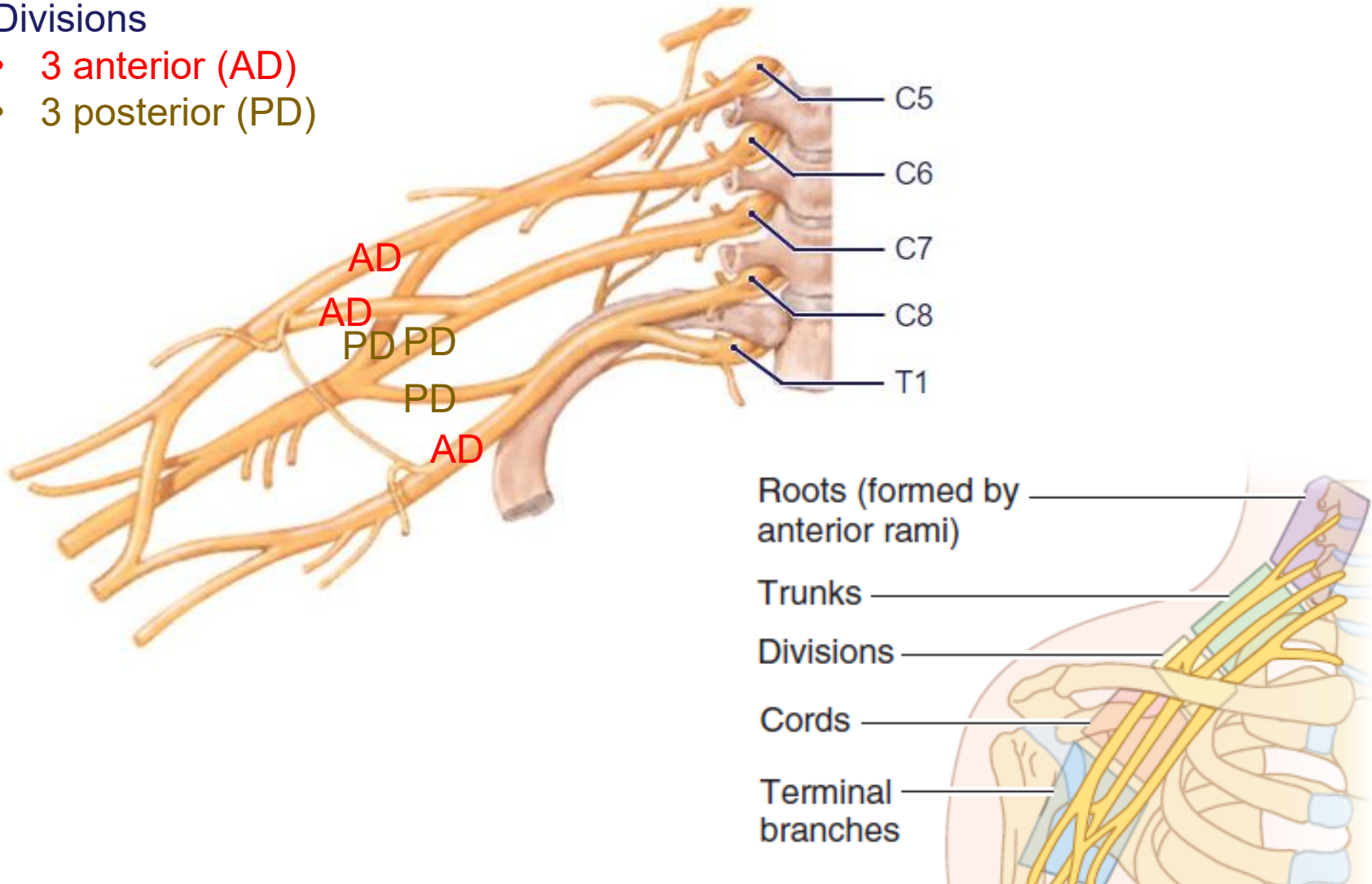
Brachial Plexus: 3 Trunks



Brachial Plexus: 6 Divisions

Divisions

- 3 anterior (AD)
- 3 posterior (PD)



Brachial Plexus: 3 Cords

Cords

- **Lateral (LC)**
- **Posterior (PC)**
- **Medial (MC)**

Lateral **pectoral**

- pectoralis major
- C5-7

Medial **pectoral**

- pectoralis major
- pectoralis minor
- C8,T1

Upper **subscapular**

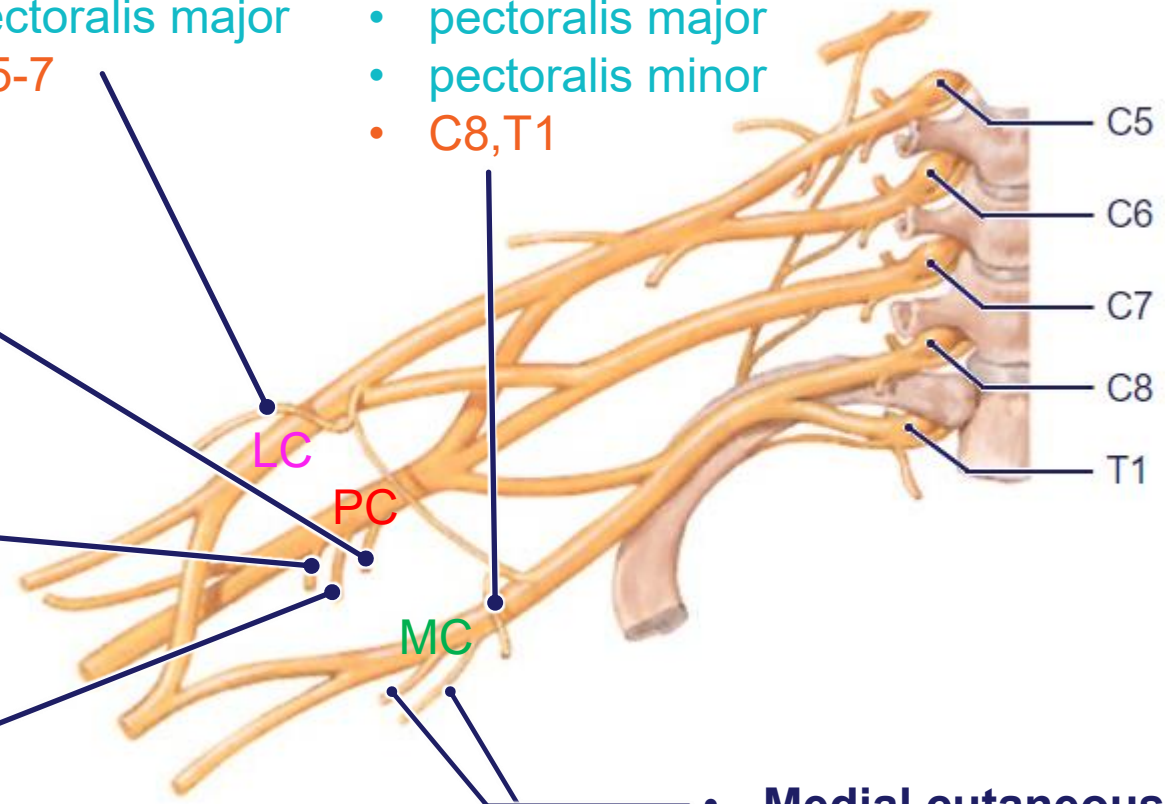
- subscapularis
- C5-7

Lower **subscapular**

- subscapularis
- teres major
- C5-7

Thoracodorsal

- Latissimus dorsi
- C6-8



- **Medial cutaneous** nerve of the arm
- **Medial cutaneous** nerve of the forearm
- C8,T1

Brachial Plexus: 5 Branches

Musculocutaneous

- C5-7

Axillary

- C5,6

Radial

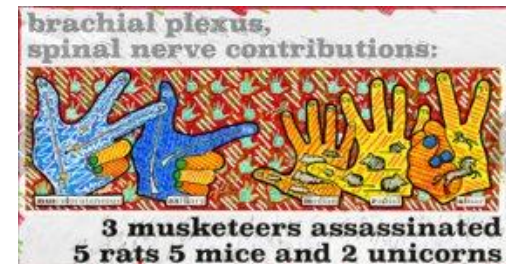
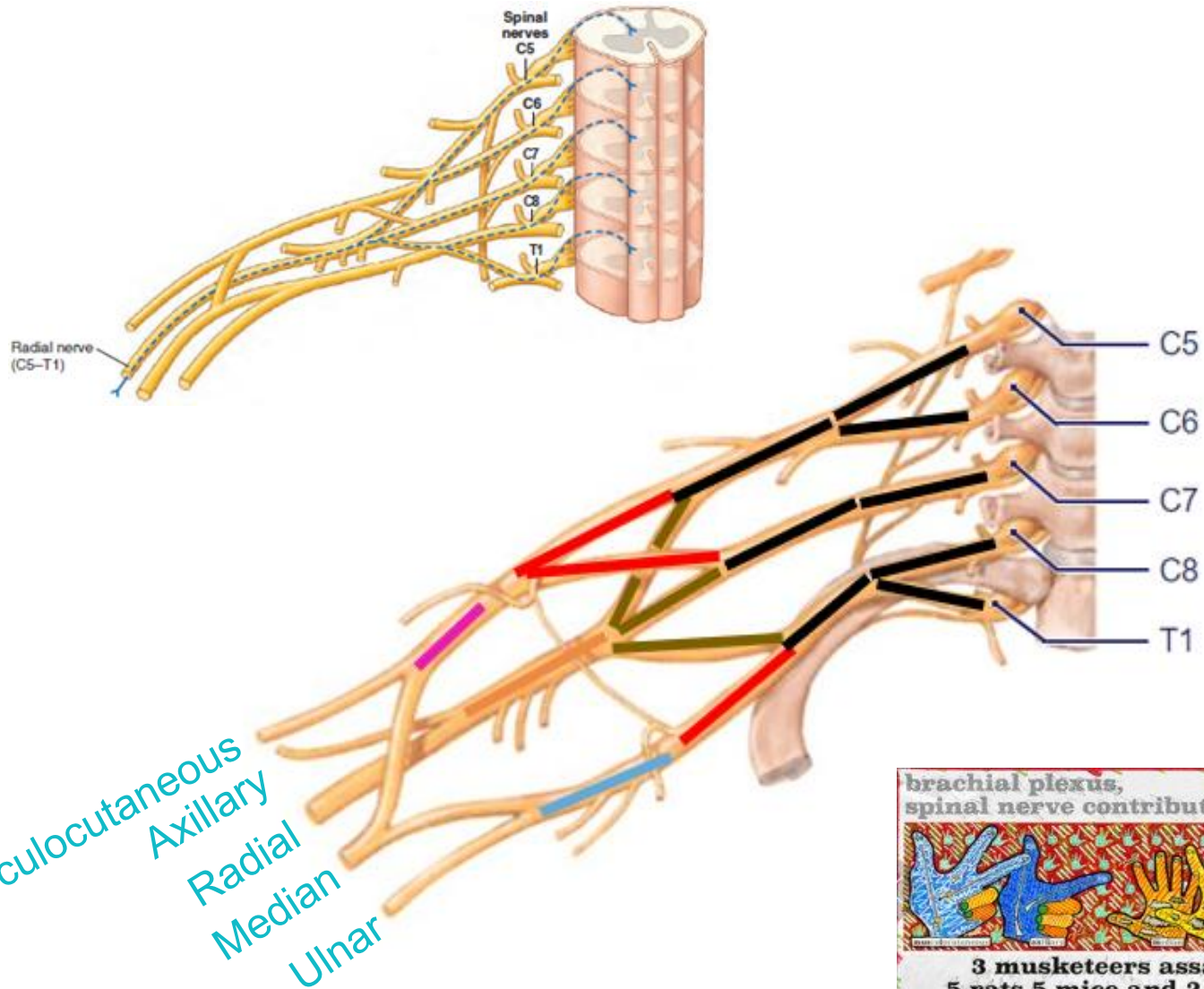
- C5-T1

Median

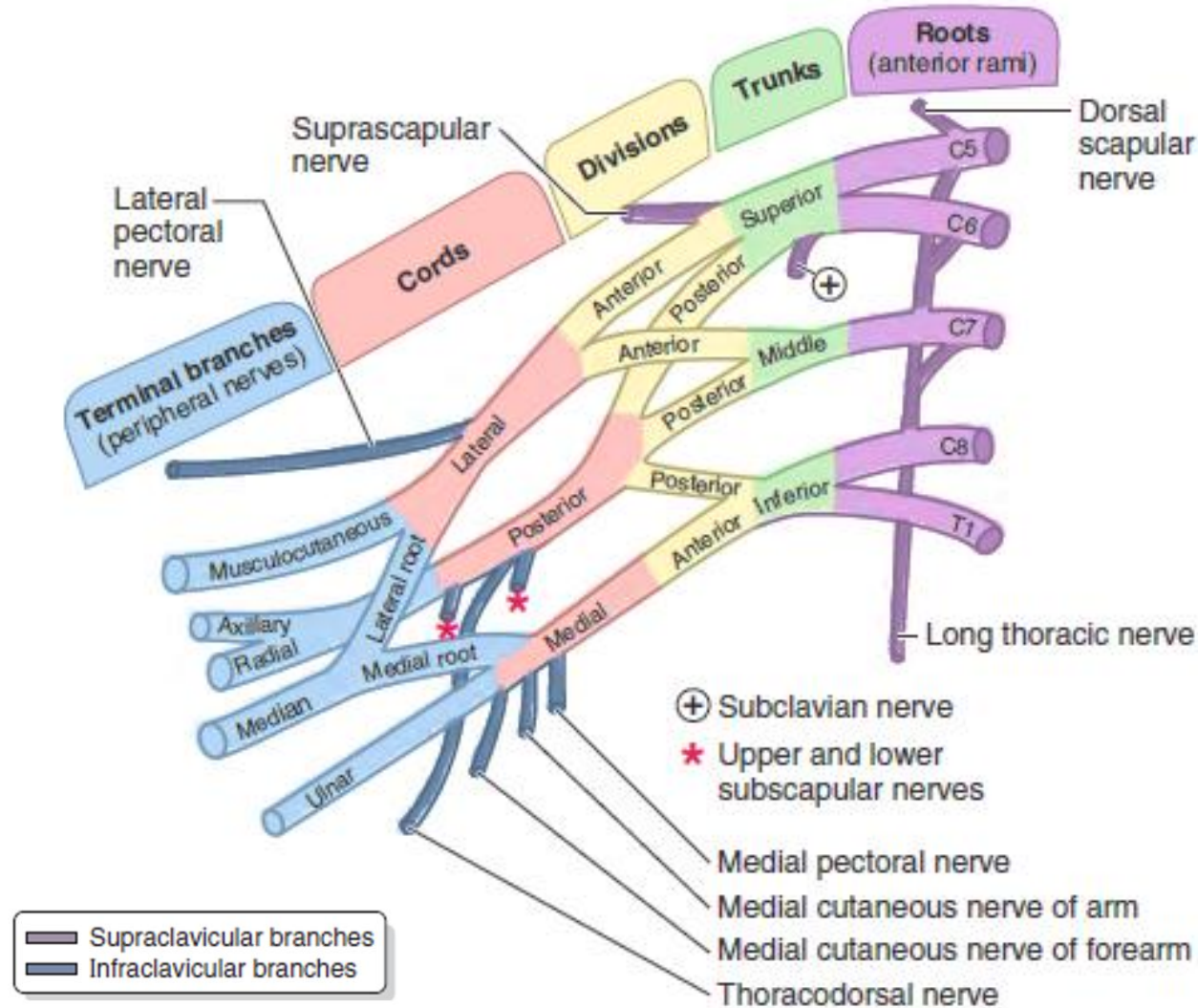
- C5-T1

Ulnar

- C8,T1

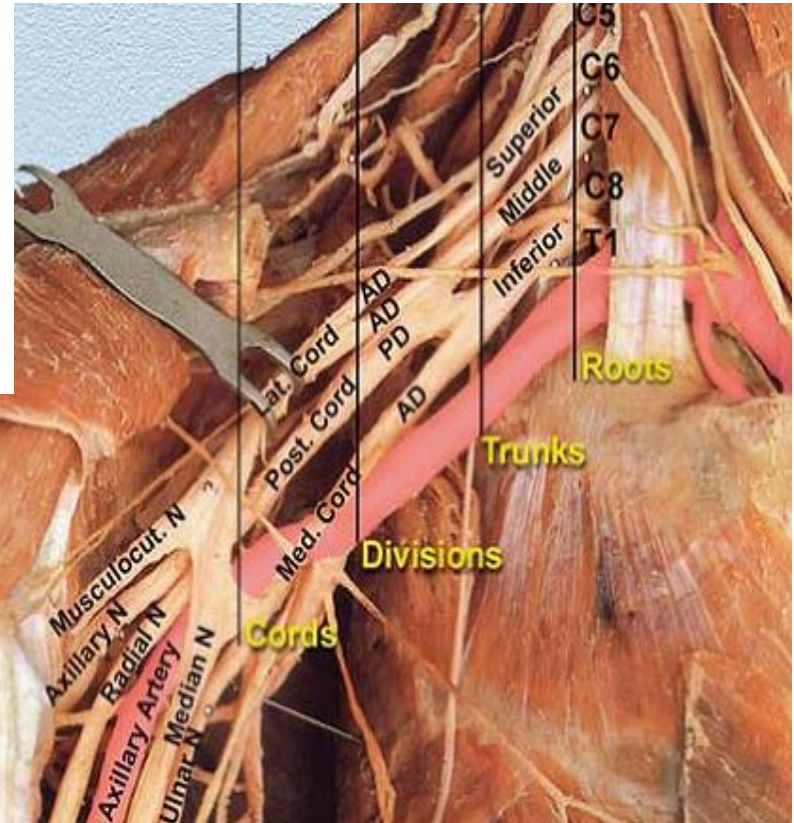
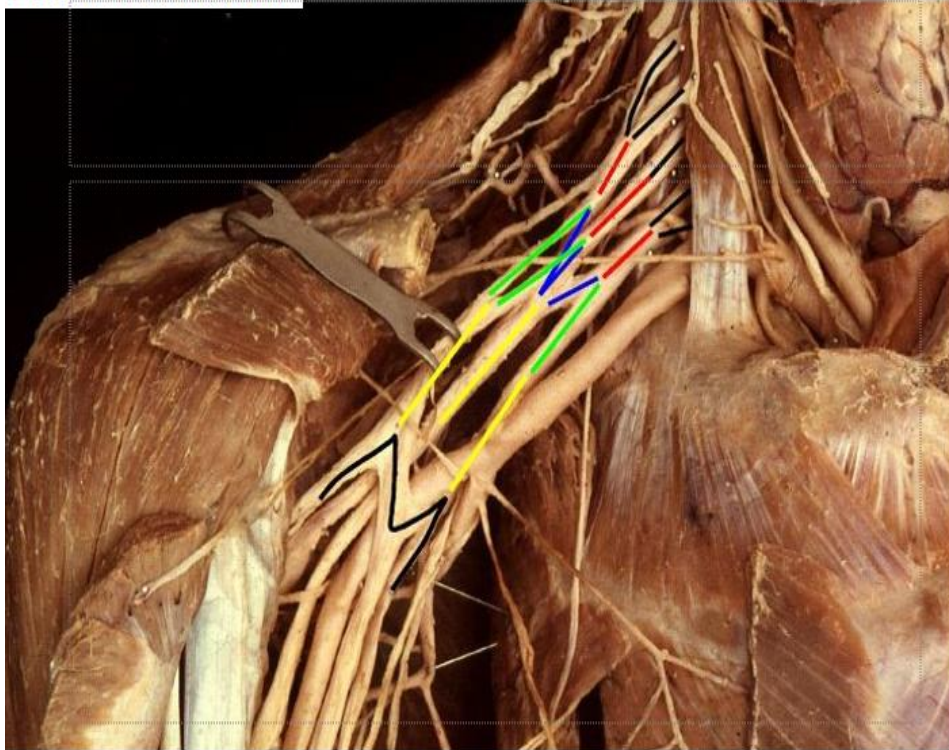


Brachial Plexus In Toto



Brachial Plexus In Situ

BRACHIAL PLEXUS



Complete the table (for self-review)

Lesion	Nerves affected	Muscles affected	Functions lost
Upper brachial plexus lesion • Erb's palsy • C5 & C6 nerve roots or upper trunk			
Lower brachial plexus lesion • Klumpke's palsy • C8 & T1 nerve roots or lower trunk			



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